

Clase 6:

Sistemas Recomendadores

Vicente Domínguez

Diplomado en Machine Learning Aplicado UC



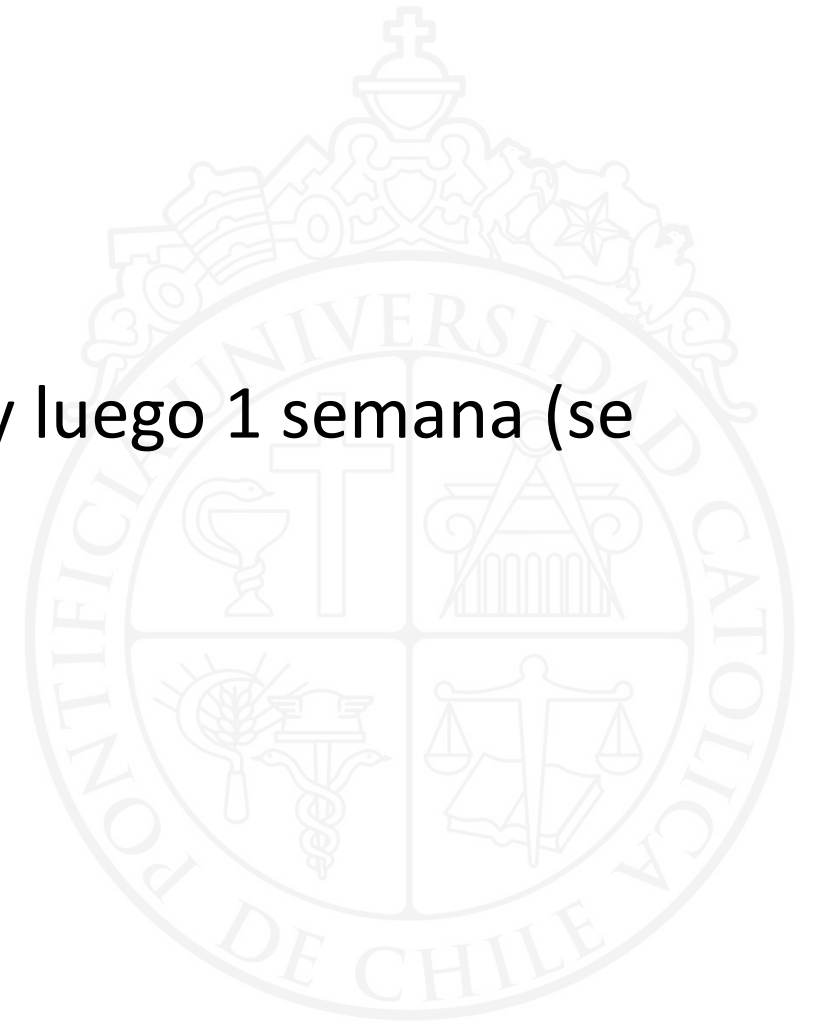
Programa del Curso

1. Introducción , Recomendación No personalizada , Filtrado Colaborativo (User y Item), Evaluación basada en métricas de error (RMSE, MAE) + **Práctico**.
2. Slope One , Recomendación basada en métodos latentes (Factorización Matricial) + Práctico.
3. Recomendación basada en feedback implícito , Optimización basada en ALS y BPR, Evaluación basada en métricas de ranking, novedad, serendipia + Práctico.
4. Recomendación basada en contenido (text mining , nlp , aplicación) + Práctico.
5. Recomendación basada en contexto y recomendación híbrida. + Práctico.
6. **Deep Learning para recomendación (texto , imágenes, multimodal, secuencial) + Práctico.**
7. Reinforcement Learning (Bandits) + Conversational Recsys + **Práctico**
8. Repaso general , dominios de aplicación e investigación reciente , recomendación basada en grafos.

Evaluación

Promedio ejercicios prácticos.

Tiempo para realizarlos. Al final de la clase y luego 1 semana (se entrega en la siguiente clase)



En esta clase

1. Repaso de recomendación contextual e híbrida.
2. Recomendación basada en deep learning.
3. Práctico recomendación de imágenes.



Repaso



Contexto

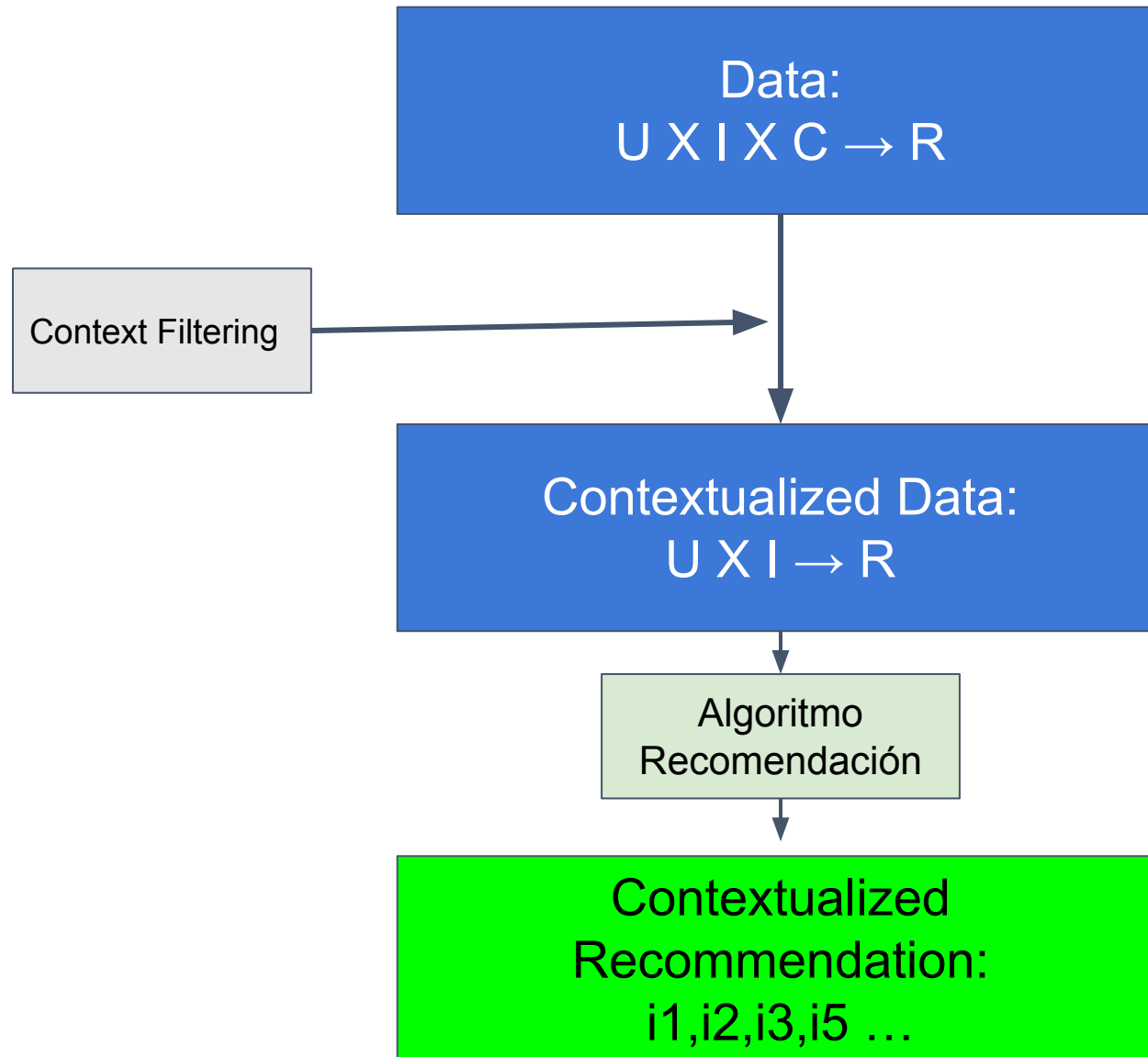
Un factor importante a considerar en las recomendaciones.



Set de datos de User-Item-Context

User	Item	Who	Where	When	Rating
U1	M1	Kids	Home	Weekend	5
U1	M2	Family	Theater	Weekend	4
U1	M3	Partner	Event	Weekday	5
U2	M1	Friends	Home	Weekend	3
U2	M2	Family	Home	Weekday	4
U3	M2	Kids	Theater	Weekday	2
U3	M3	Partner	Home	Weekend	1
U2	M3	Partner	Home	Weekday	?

Contextual pre-filtering



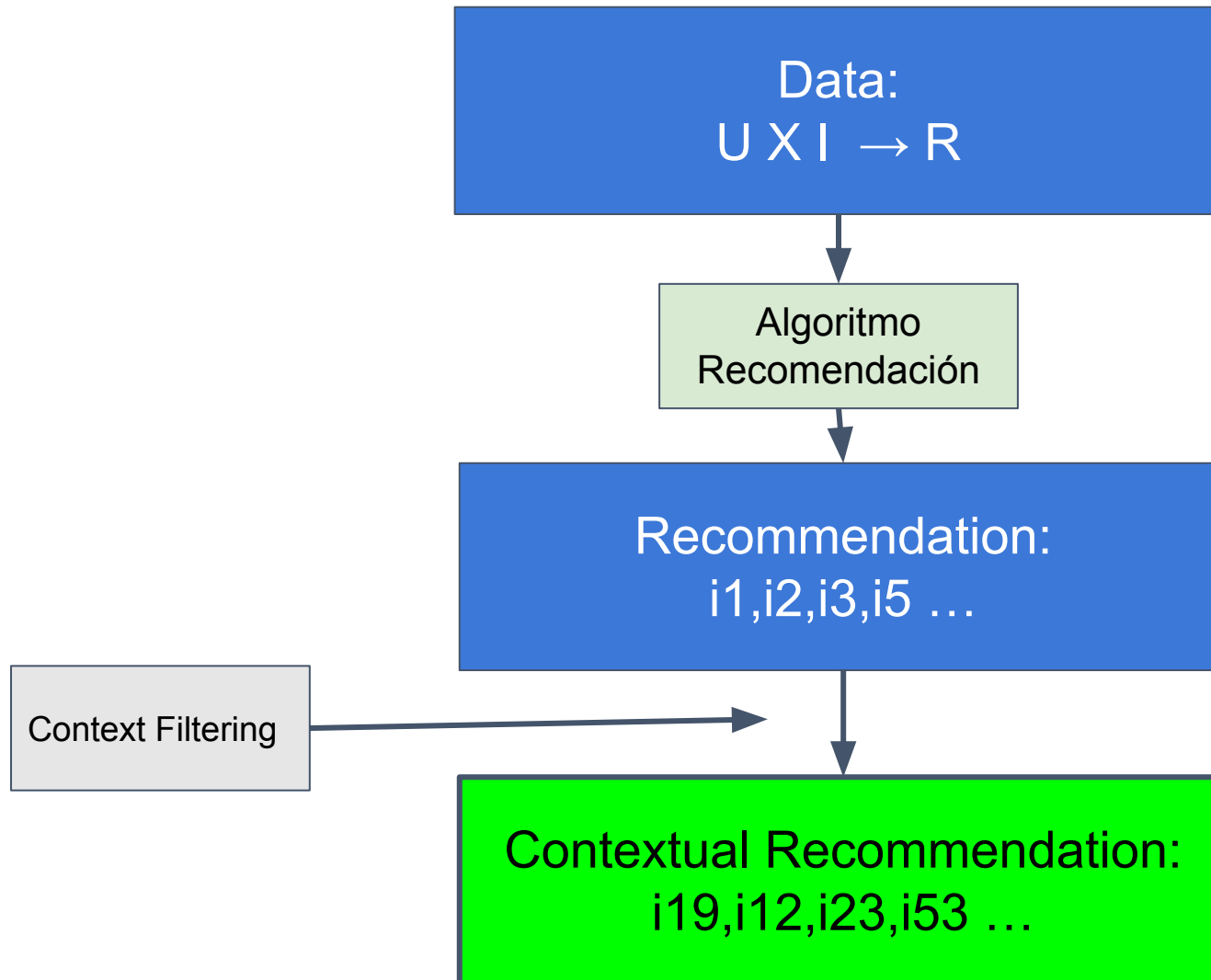
ERROR COMÚN:

ESCOGER UN CONTEXTO DEMASIADO ESPECÍFICO.

BUEN CONTEXTO:

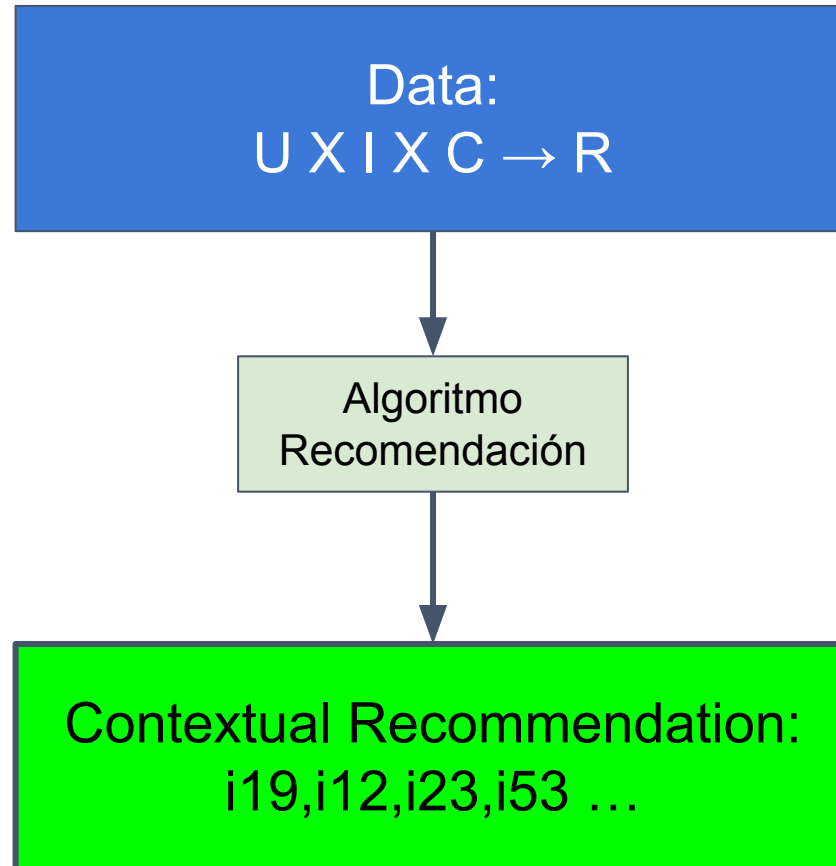
- DIA / NOCHE
- INVIERNO
- LOCAL DE MACUL

Contextual post-filtering



IGNORAMOS EL
CONTEXTO ANTES DE
HACER LA
RECOMENDACION

Contextual modeling



APRENDE VECTORES:

- USUARIO
- ITEM
- CONTEXTO

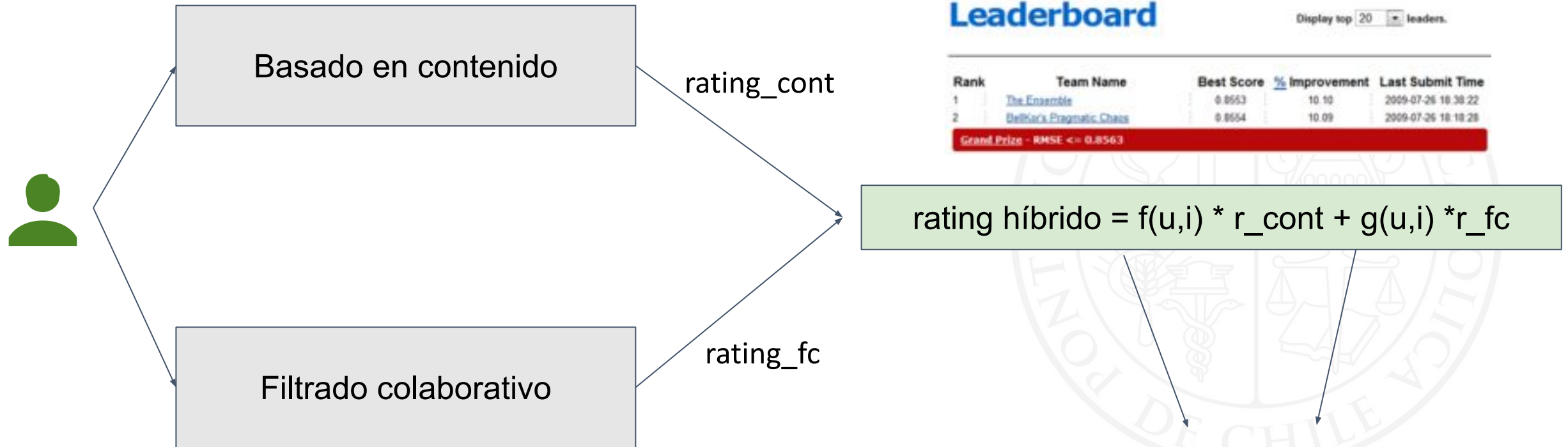
INCORPORA EL CONTEXTO
EN EL MODELO

Recomendación híbrida

1. Votación entre distintos algoritmos de recomendación.
2. Utilizar heurísticas para decidir qué algoritmo de recomendación utilizar.
3. Pesos (aprendidos o como meta-parámetro) asignado a cada recomendador.
4. Feature ensemble: agrega features adicionales a un algoritmo de recomendación.

Ensamble basado en pesos entregados por un método de aprendizaje

Netflix Prize uso un ensemble de múltiples métodos (FM y RBM)



pesos ahora son funciones $f(u,i)$ y $g(u,i)$
que aprenden los pesos para cada
recomendador.

Factorization machines [1/2]

Matrix Factorization
Training Data

	i_1	i_2	i_3
u_1	2	4	
u_2		1	
u_3	3		5



Factorization Machine
Training Data

	u_1	u_2	u_3	i_1	i_2	i_3	a_1	a_2	y	
x_1	1	0	0	1	0	0	2.0	0.0	2	Observed Ratings
x_2	1	0	0	0	1	0	1.5	0.5	4	
x_3	0	1	0	0	1	0	0.0	1.0	1	
x_4	0	0	1	1	0	0	0.3	0.7	3	
x_5	0	0	1	0	0	1	3.2	1.7	5	
	Users			Items			Auxiliary Features			

Factorization machines [2/2]

$$\hat{y}(\mathbf{x}) := w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=i+1}^n \langle \mathbf{v}_i, \mathbf{v}_j \rangle x_i x_j$$

Coeficientes de
regresión de la
iesima variable

Factorización
Matricial
User - Item

Interacción
entre features
 x_i y x_j

Reflexión

Hasta ahora hemos generado recomendación basada en datos tabulares y texto.

¿Cómo podemos incorporar deep learning a recomendación?

¿Cómo introducimos información de imágenes o de data no estructurada (ej. audio, video)?

En esta clase

1. Recomendación utilizando modelos de deep learning.
2. Práctico recomendación de imágenes.



Recomendación utilizando contenido de imágenes

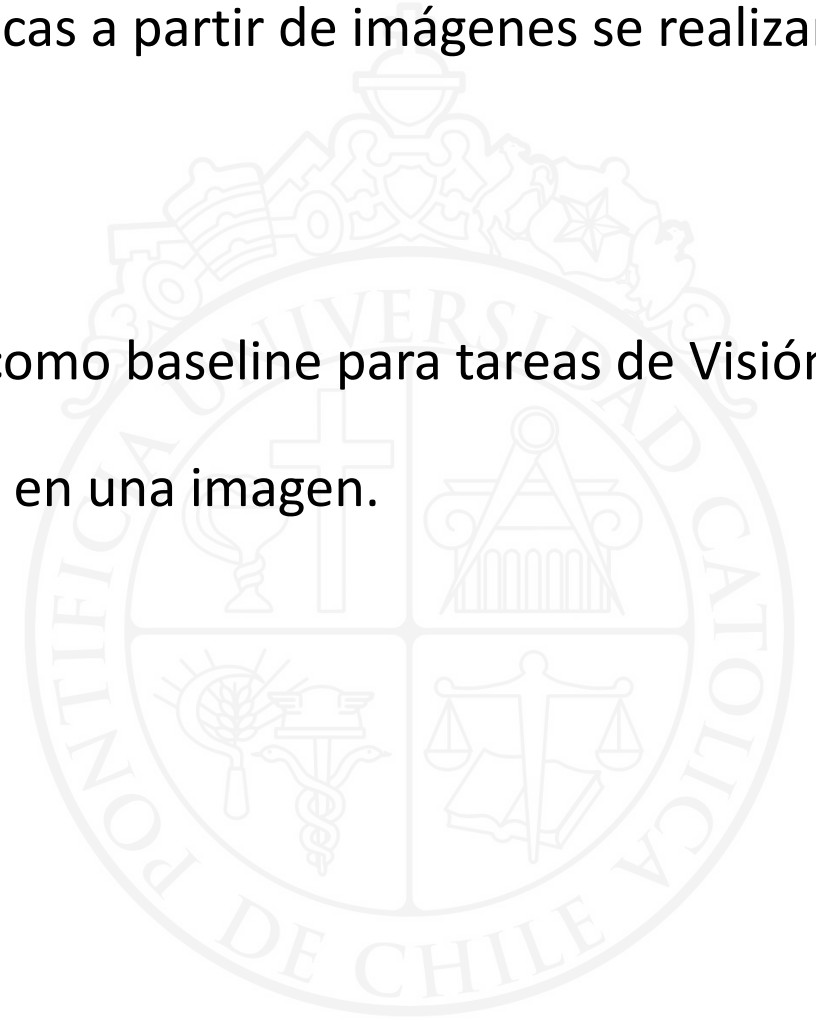


Recomendación visual basada en contenido

En los modelos tradicionales de extracción de características a partir de imágenes se realizan con diferentes técnicas.

Algunas de ellas:

- Local Binary Patterns: método semi-manual usado como baseline para tareas de Visión por Computador.
- Obtiene un histograma de 59 patrones encontrados en una imagen.
- En base a 7 métricas.



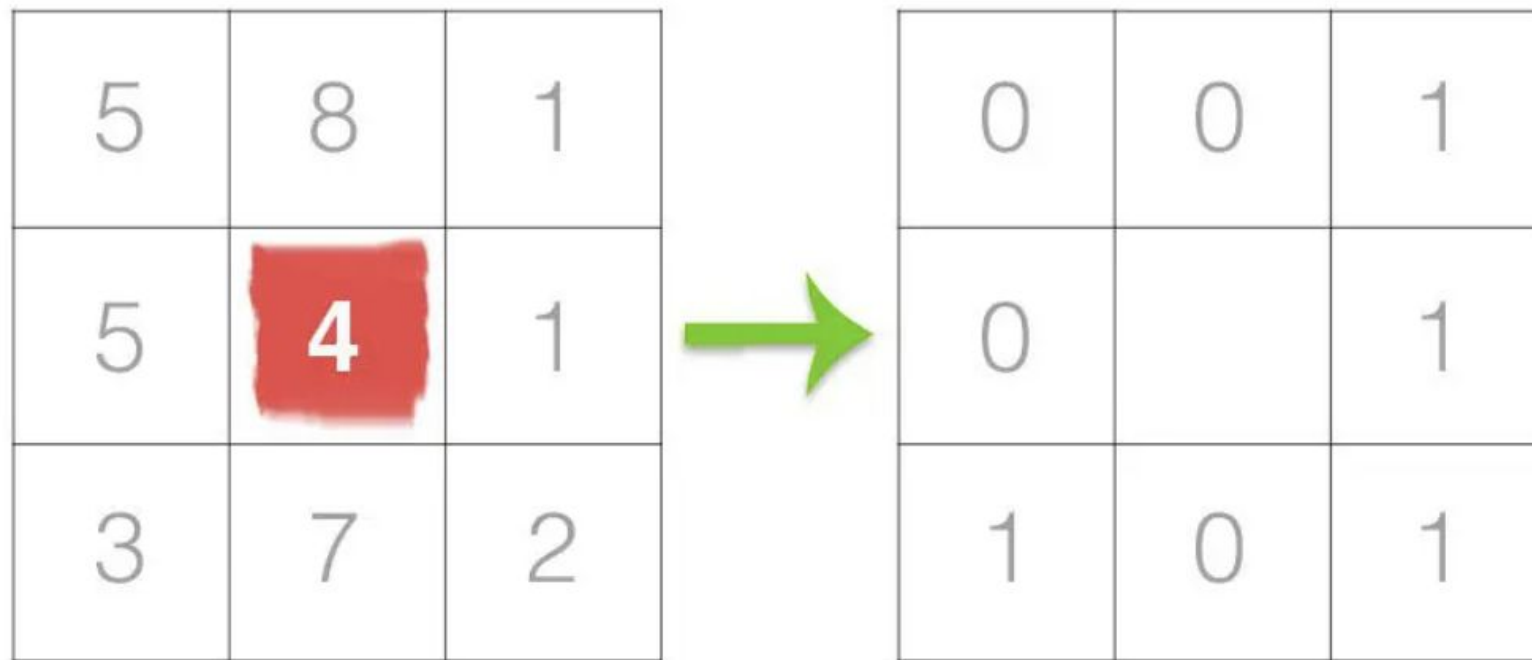


Figure 1: The first step in constructing a LBP is to take the 8 pixel neighborhood surrounding a center pixel and threshold it to construct a set of 8 binary digits.

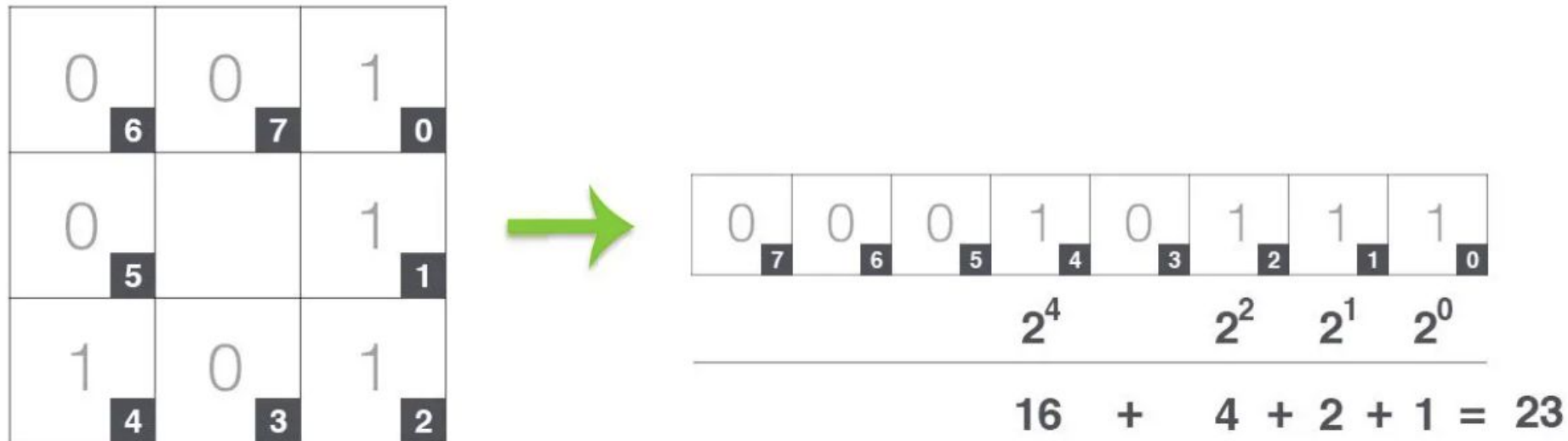


Figure 2: Taking the 8-bit binary neighborhood of the center pixel and converting it into a decimal representation. (Thanks to Bikramjot of [Hanzra Tech](#) for the inspiration on this visualization!)

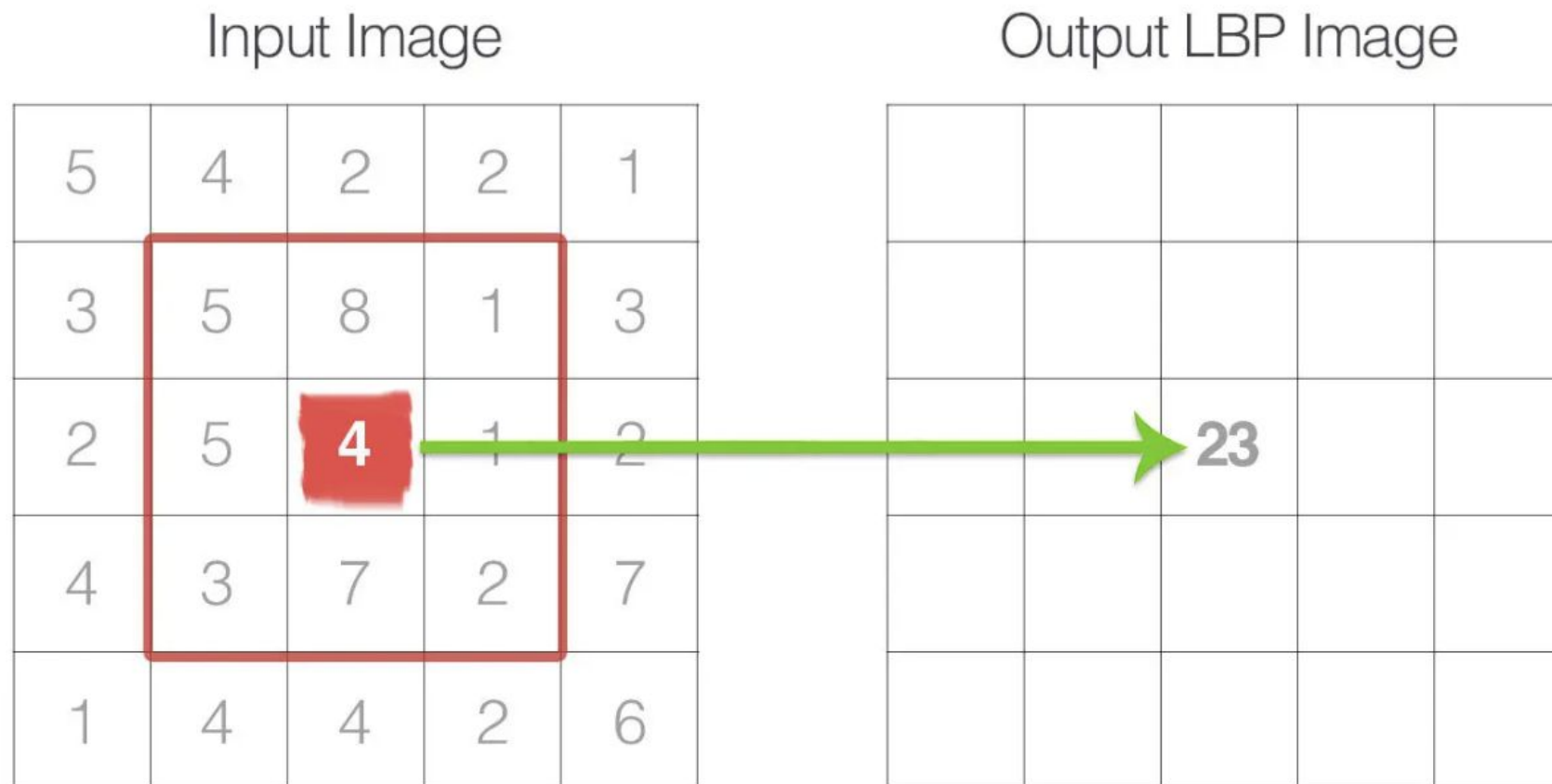
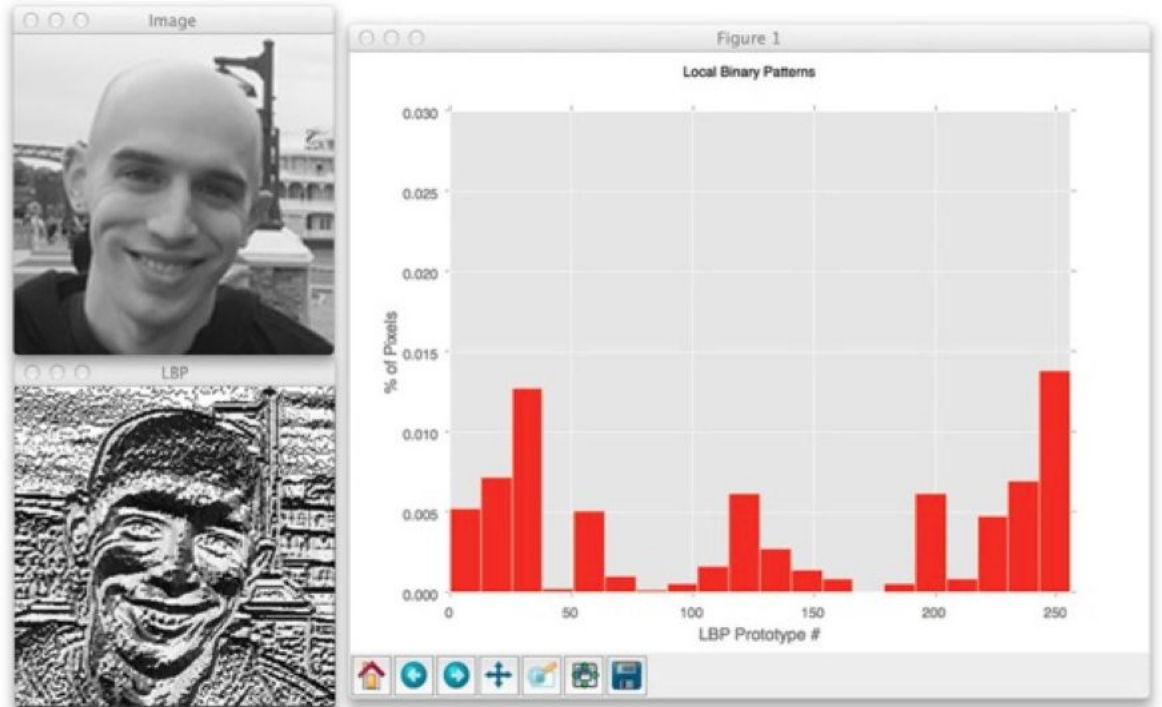


Figure 3: The calculated LBP value is then stored in an output array with the same width and height as the original image.

LBP: LOCAL BINARY PATTERNS

- Finalmente se calcula un histograma que tabula el número de ocasiones en que cada patron LBP ocurrió.
- Podemos pensar en este histograma como un vector de features.



Fuente: <https://www.pyimagesearch.com/2015/12/07/local-binary-patterns-with-python-opencv/>

Características visuales atractivas

San Pedro y Sierdorfer (2009) estudiaron características para caracterizar imágenes por su atractivo visual:

- Brightness (brillo)
- Saturación (saturation)
- Sharpness (nitidez)
- RMS-contrast (contraste RMS)
- Colorfulness (colorido)
- Naturalness (naturalidad)
- Entropy (entropía)

Jose San Pedro and Stefan Siersdorfer. 2009. Ranking and classifying attractiveness of photos in folksonomies. In Proceedings of the 18th international conference on World wide web (WWW '09).



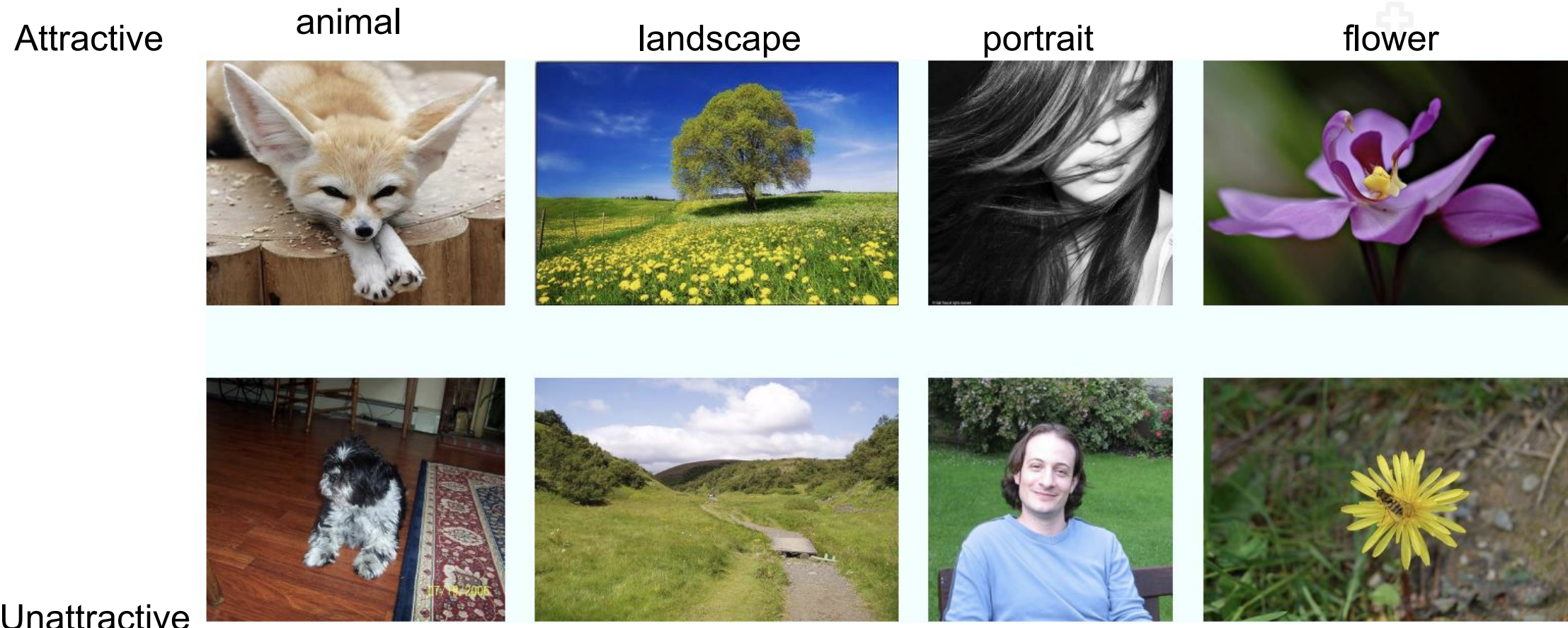
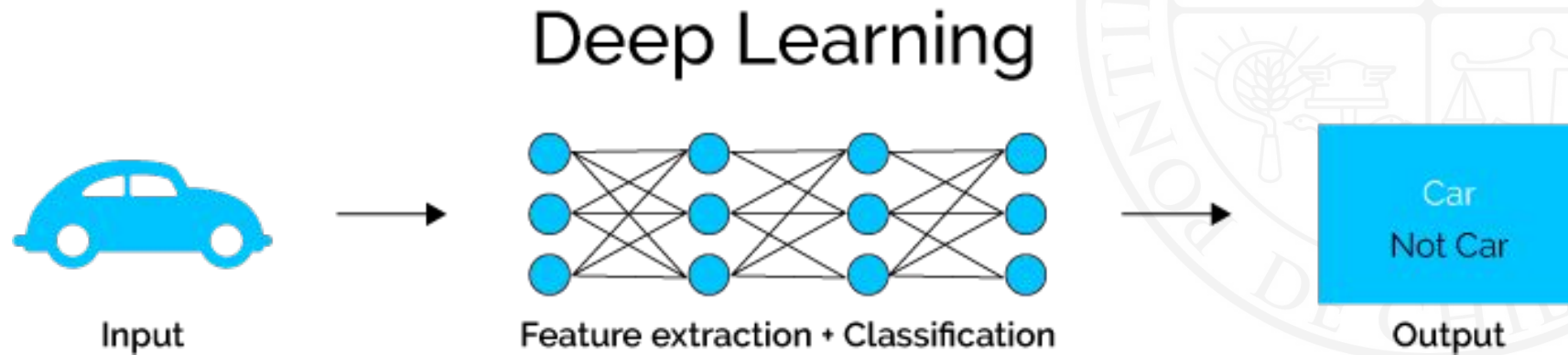
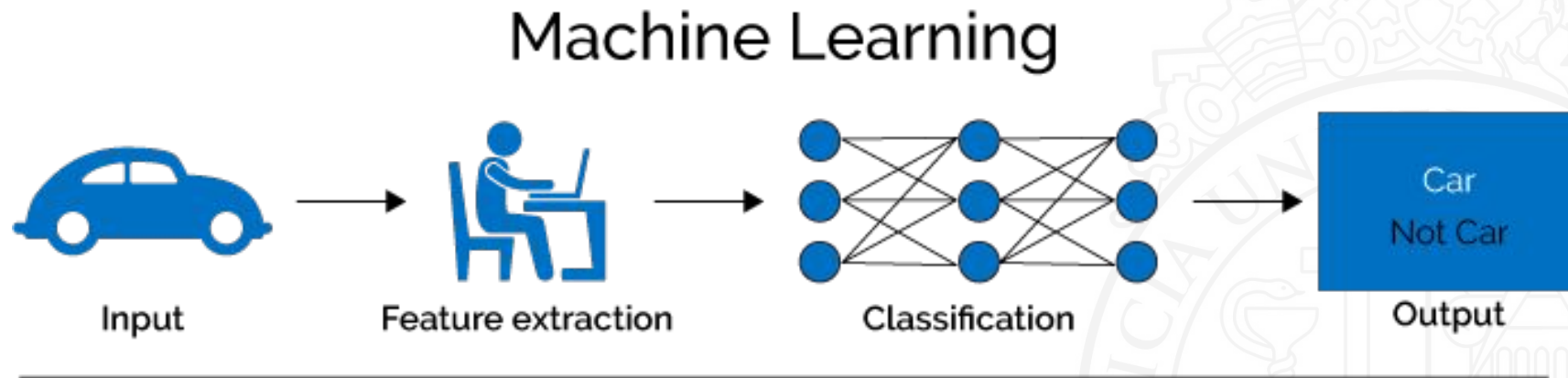


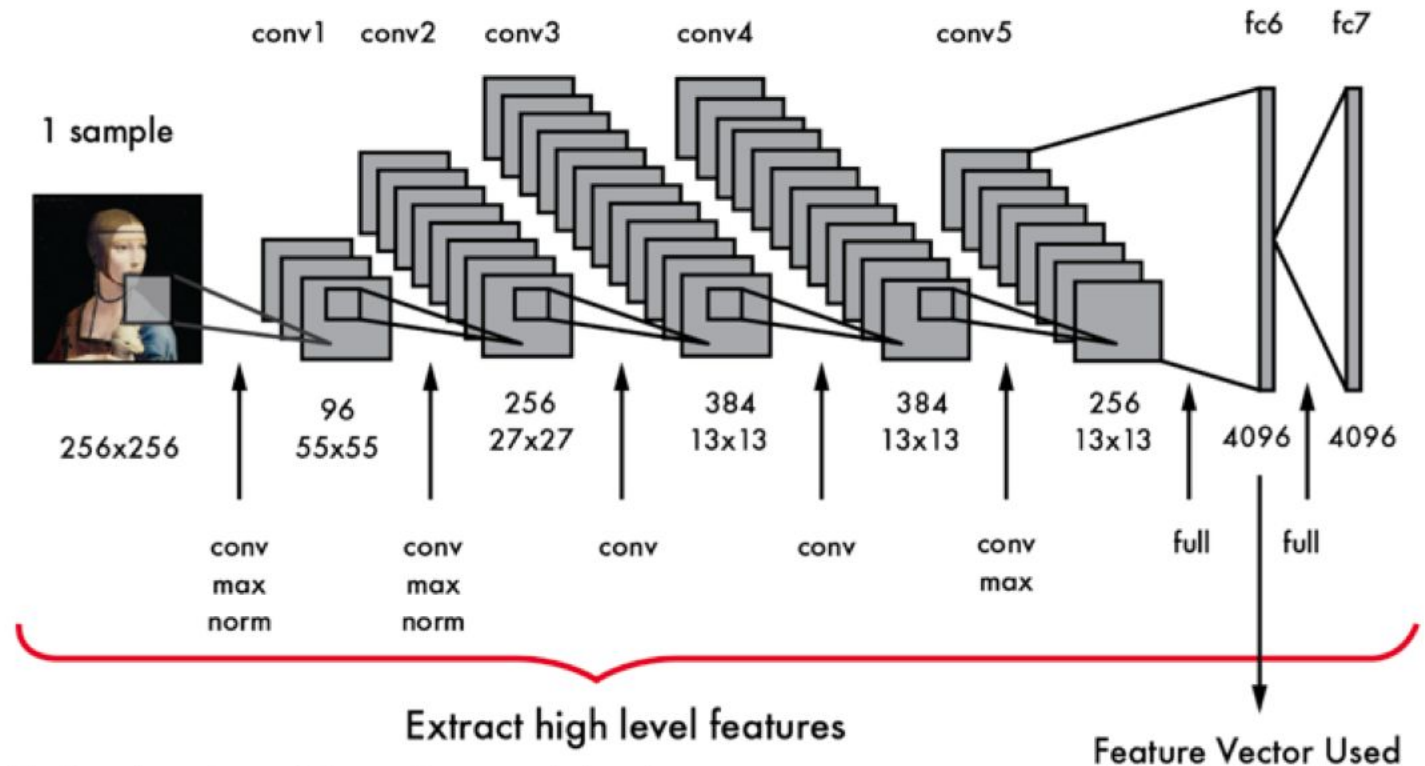
Figure 1: “Attractive” (upper row) vs. “Unattractive” (lower row) images: Each column represents the same semantic concept (animal, landscape, portrait, flower) but differences in appeal-related visual attributes.

¿Por qué deep learning?



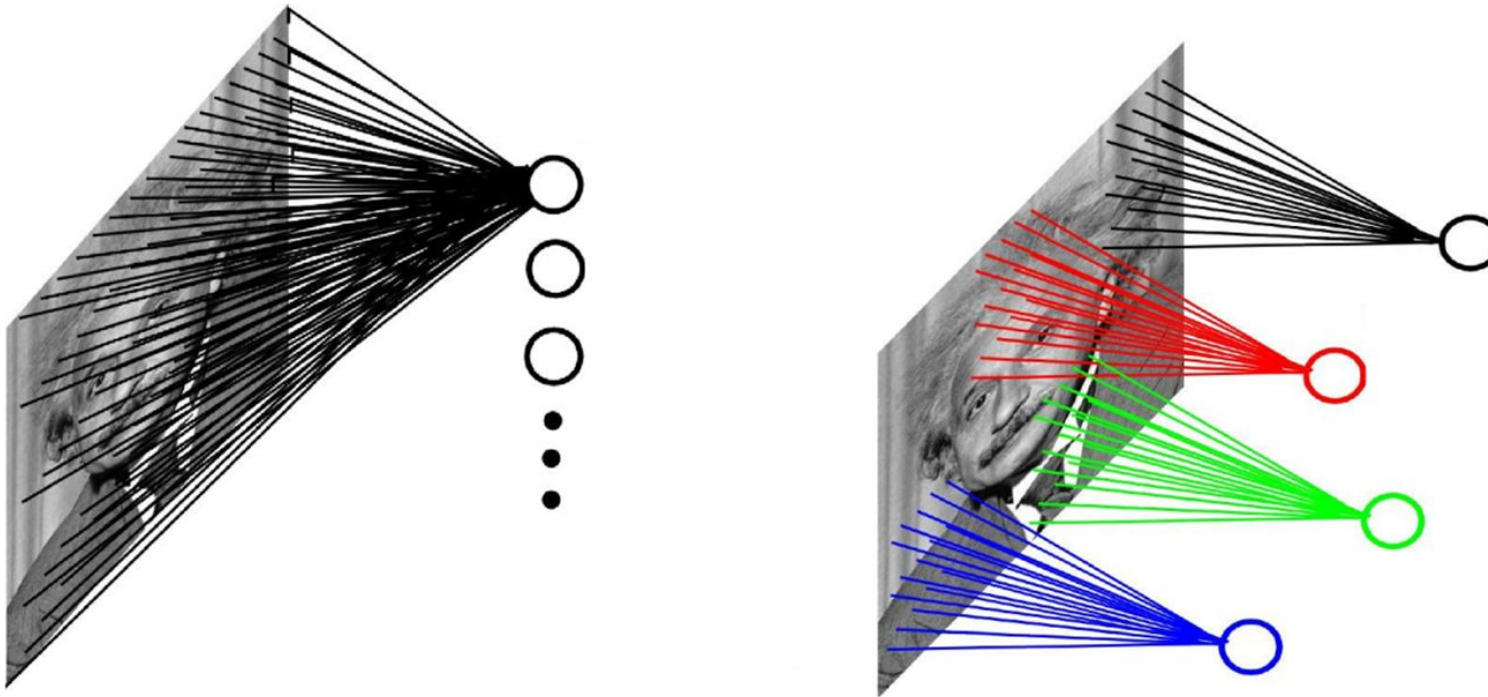
Features manuales versus Deep Learning

- Con DL podemos usar features aprendidas automáticamente con una red neuronal pre-entrenada para otra tarea: clasificación de objetos del dataset Imagenet.



Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*.

¿Cómo podemos entonces capturar esta información de manera automática sin feature engineering?



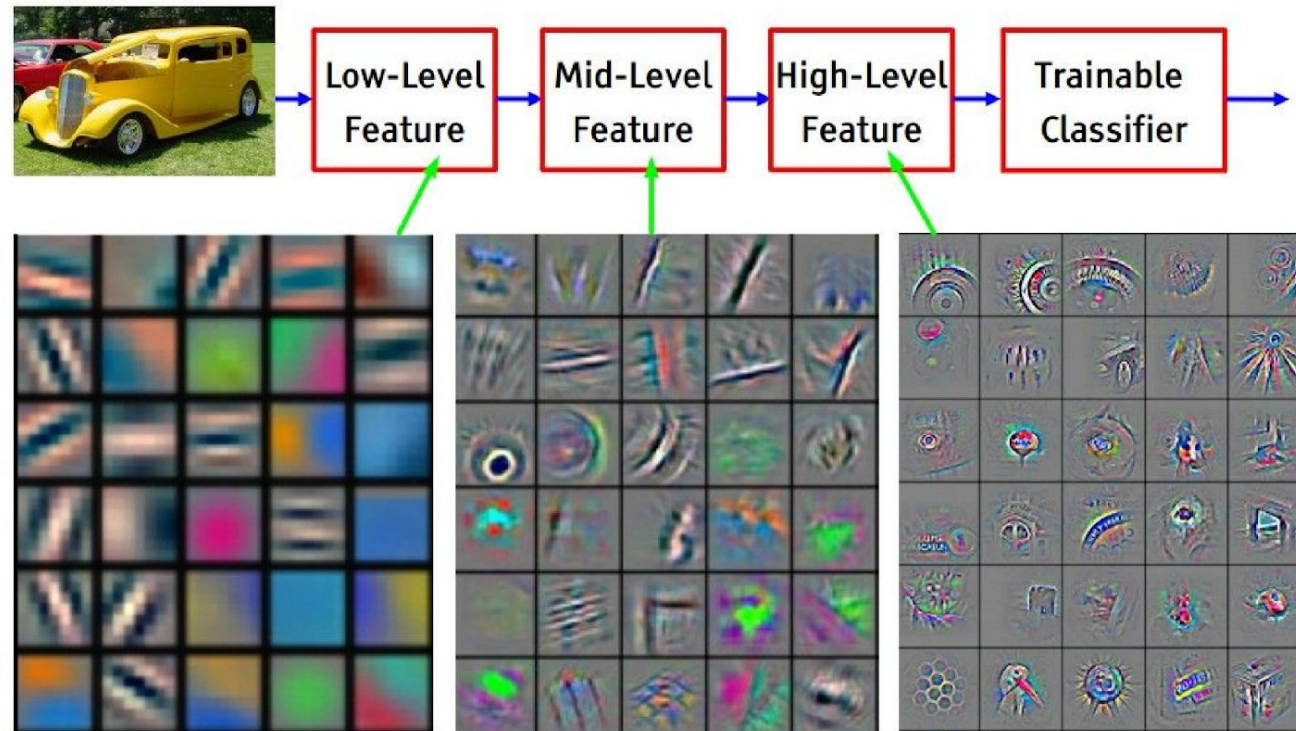
Neuronas capaces de barrer la entrada proveen una reducción significativa en el número de parámetros (no depende de la conectividad)

Además, dado su tamaño, permiten capturar patrones locales en cualquier posición.

¿Cómo podemos entonces capturar esta información de manera automática sin feature engineering?

Preview

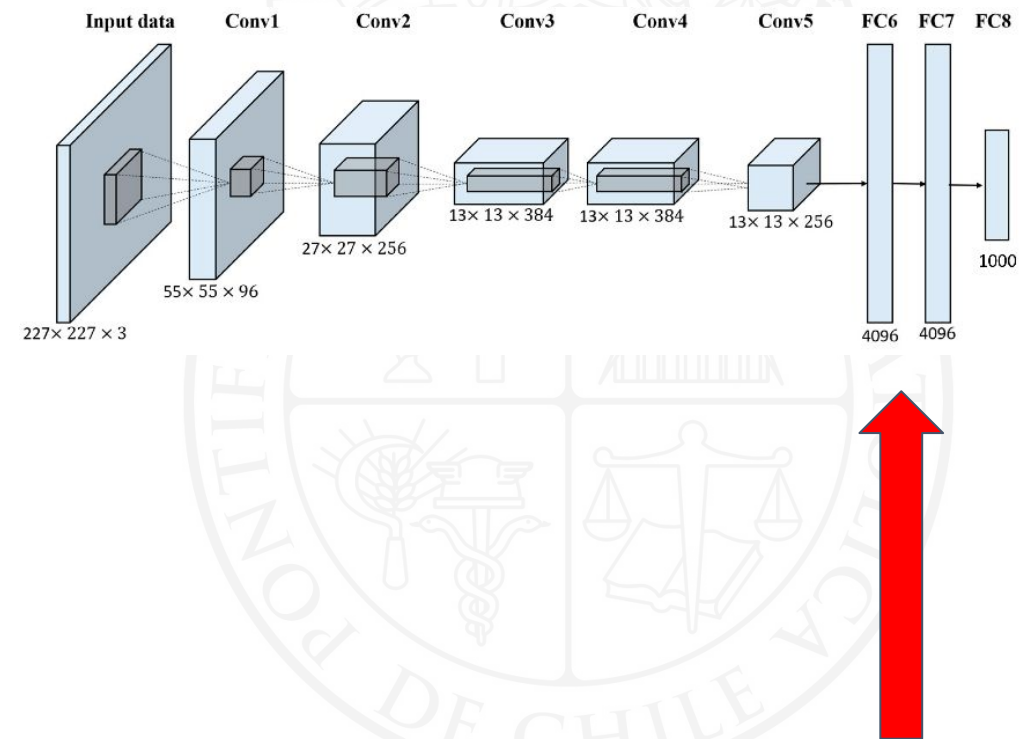
[From recent Yann LeCun slides]



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Redes neuronales convolucionales profundas

- Están formadas por capas convolucionales, de pooling y fully connected (MLP).
- Sustituyen la mayoría de las capas densas (fully connected) de los MLP, por capas convolucionales.
- Disminuyen drásticamente el número de parámetros, en comparación con los MLP.
- Sustentadas en bases biológicas (neuronas reales)

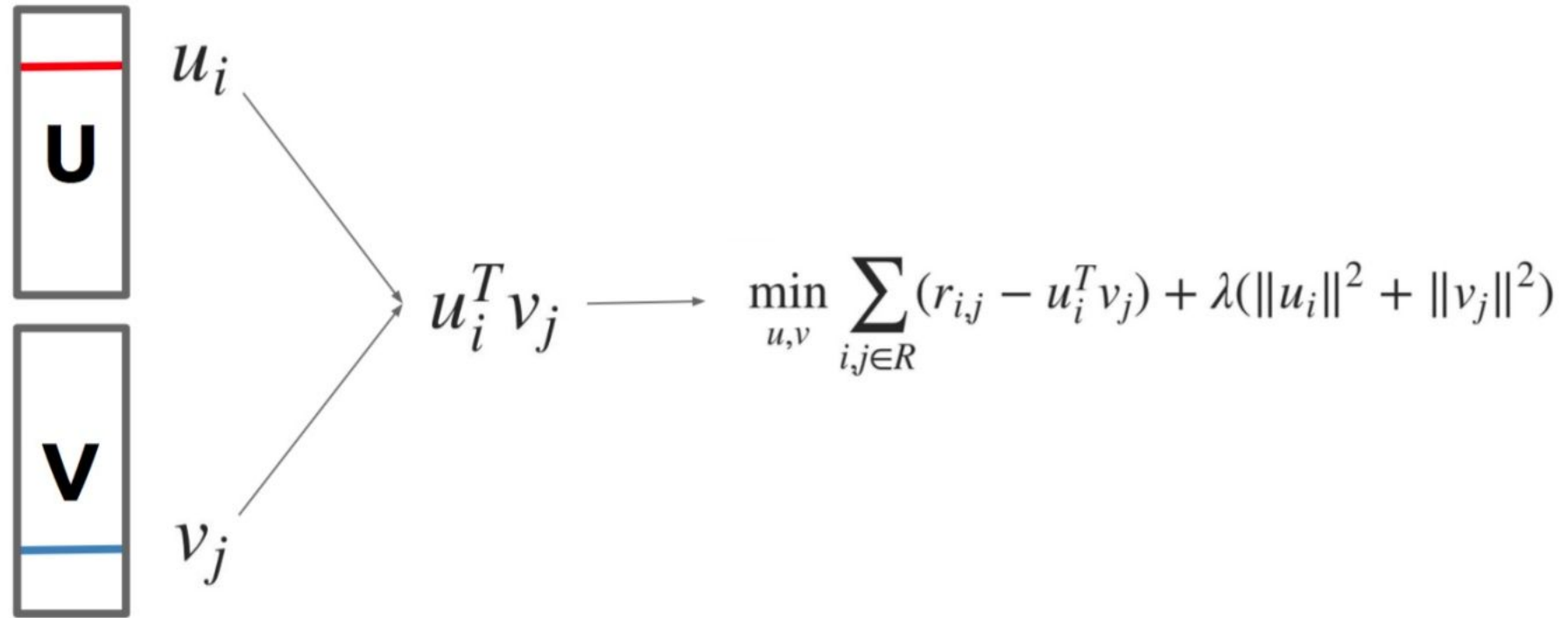


EMBEDDING

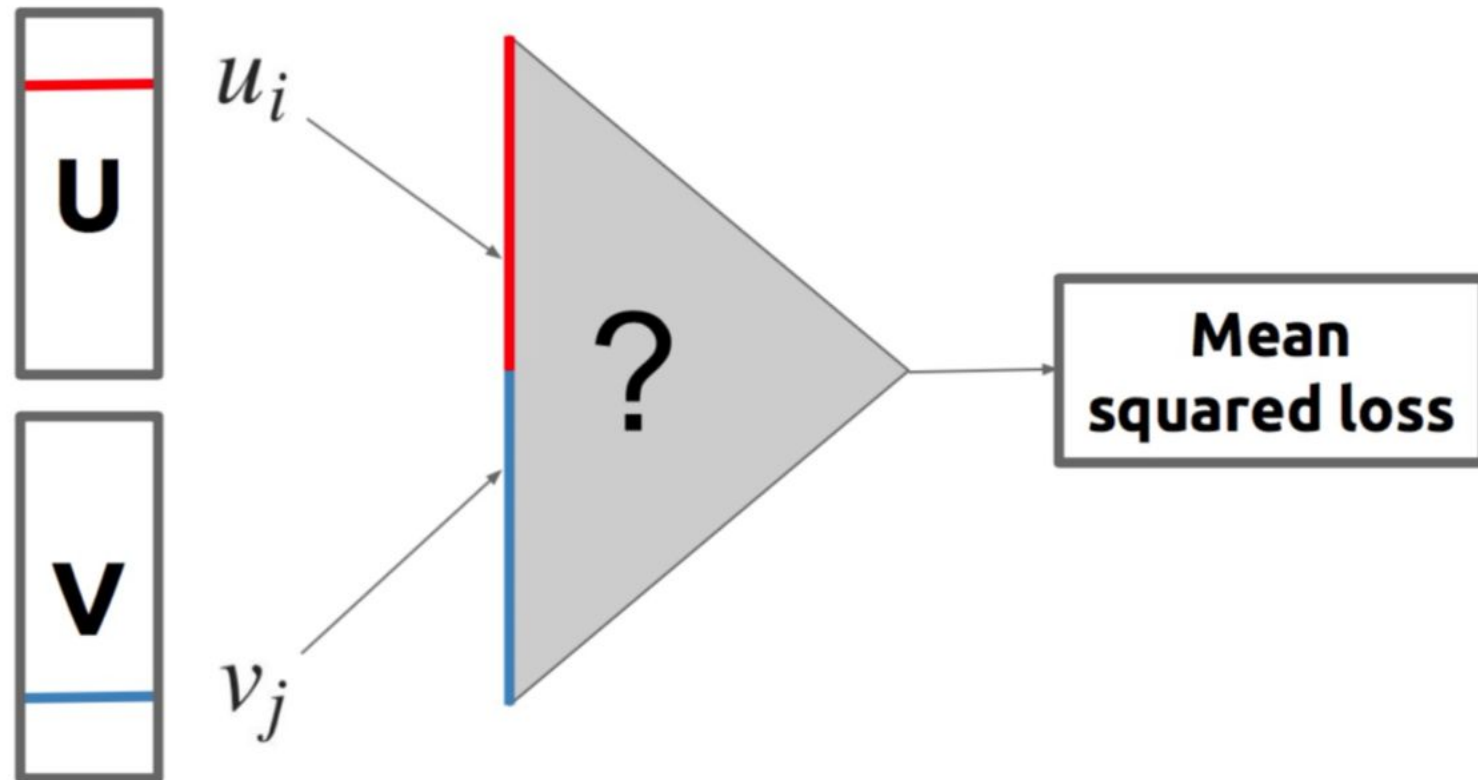
Sistemas de recomendación utilizando deep learning



Factorización Matricial con Redes Neuronales



Factorización Matricial con Redes Neuronales



Factorización Matricial con Redes Neuronales

- Redes neuronales y factorización matricial son similares
 - Uso de embeddings
 - Pérdida mínimos cuadrados
 - Óptimo con descenso de gradiente
- Una red neuronal puede aprender más combinaciones que el producto punto
- Una red neuronal requiere muchos datos para aprender patrones e interacciones
- El uso de redes neuronales no es mejor que una buena factorización matricial... por ahora...

Las redes neuronales permiten capturar más información que una factorización matricial como información de texto, imágenes, música, etc.

Redes Neuronales para Factorización Matricial

- <https://arxiv.org/pdf/1907.06902.pdf>

Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches

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Paolo Cremonesi
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Dietmar Jannach
University of Klagenfurt, Austria
dietmar.jannach@aau.at

- Si solo usamos datos de interacciones, puede que las redes neuronales no nos ofrezcan tanto beneficio.

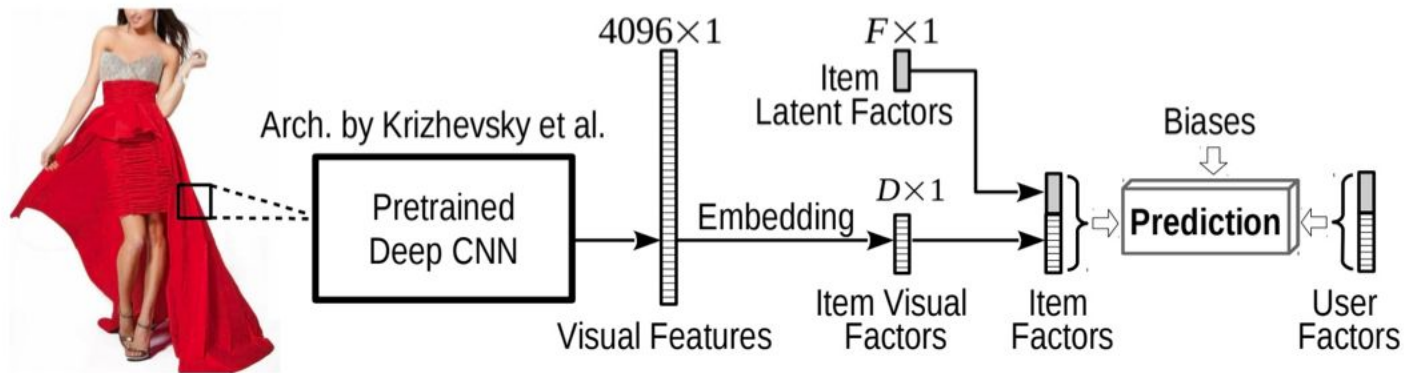
Ventajas de las Redes Neuronales

- Información de texto
 - Ejemplo: descripción de productos, comentarios de usuarios
 - Extracción: RNN
 - Aplicaciones: noticias, libros, e-commerce
- Imágenes
 - Ejemplo: foto de productos, thumbnail de videos
 - Extracción: CNN
 - Aplicaciones: moda, video
- Música / audio
 - Ejemplo: spotify
 - Extracción: CNN y RNN
 - Aplicación: música



Imágenes en recomendaciones

- [VBPR: Visual Bayesian Personalized Ranking from Implicit Feedback](#) [He and McAuley, 2016]
- Utiliza CNN para obtener características visuales
- Utilizan embeddings de usuarios e ítems
- Pérdida BPR (pairwise)



Imágenes en Recomendaciones

Table 3: AUC on the test set $\mathcal{T}$ (#factors = 20). The best performing method on each dataset is boldfaced.

Dataset	Setting	(a) RAND	(b) MP	(c) IBR	(d) MM-MF	(e) BPR-MF	(f) VBPR	improvement f vs. best f vs. e	
<i>Amazon Women</i>	All Items	0.4997	0.5772	0.7163	0.7127	0.7020	0.7834	9.4%	11.6%
	<i>Cold Start</i>	0.5031	0.3159	0.6673	0.5489	0.5281	0.6813	2.1%	29.0%
<i>Amazon Men</i>	All Items	0.4992	0.5726	0.7185	0.7179	0.7100	0.7841	9.1%	10.4%
	<i>Cold Start</i>	0.4986	0.3214	0.6787	0.5666	0.5512	0.6898	1.6%	25.1%
<i>Amazon Phones</i>	All Items	0.5063	0.7163	0.7397	0.7956	0.7918	0.8052	1.2%	1.7%
	<i>Cold Start</i>	0.5014	0.3393	0.6319	0.5570	0.5346	0.6056	-4.2%	13.3%
<i>Tradesy.com</i>	All Items	0.5003	0.5085	N/A	0.6097	0.6198	0.7829	26.3%	26.3%
	<i>Cold Start</i>	0.4972	0.3721	N/A	0.5172	0.5241	0.7594	44.9%	44.9%

RANDOM (RAND), MOST POPULAR (MP) , IMAGE-BASED RECOMMENDATION (IBR) , PAIRWISE MF (MM-MF) , BPR-MF y VBPR

Imágenes en Recomendaciones



Combinación de dos Modelos

Wide & Deep Learning for Recommender Systems [Cheng et al., 2016]

- **Modelo Profundo**
 - Utiliza embeddings
 - Se encarga de generalizar el modelo
 - **Modelo Lineal**
 - Utiliza embeddings
 - Utiliza características de los productos
 - Memoriza patrones sobre los ítems
- modelos de deep learning se sobre-ajustan a los datos, generalizan mal.
 - modelos lineales son muy simples.
- **combinar ambos approaches.**

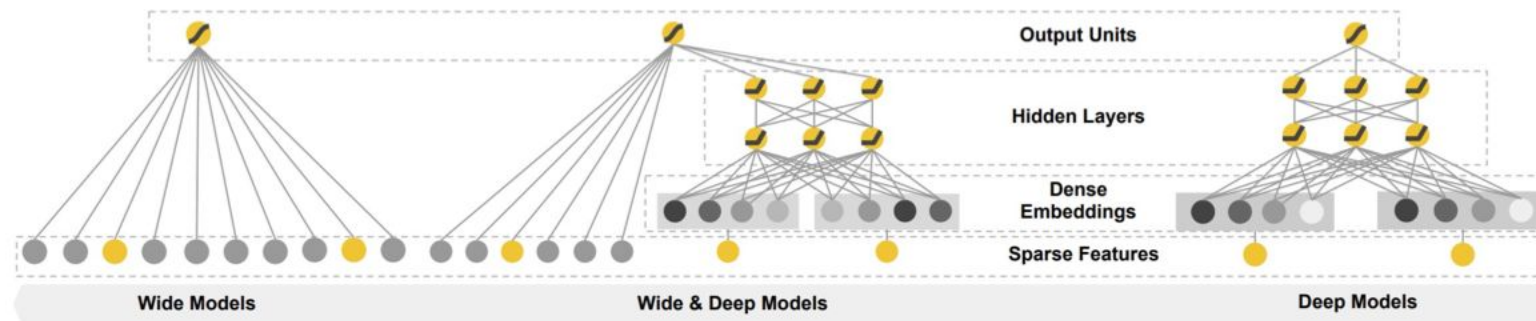


Figure 1: The spectrum of Wide & Deep models.

Combinación de dos Modelos

- Experimento del modelo implementada en Google Play Store

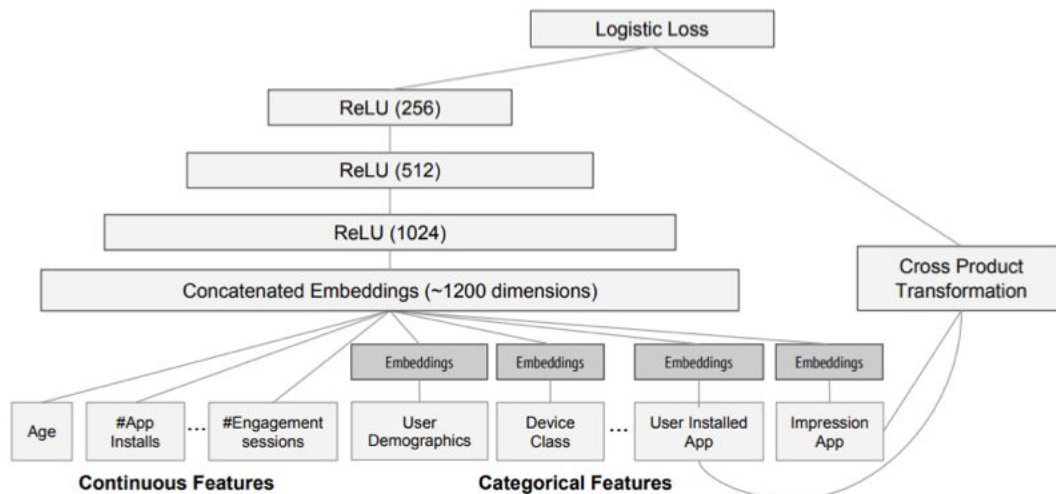


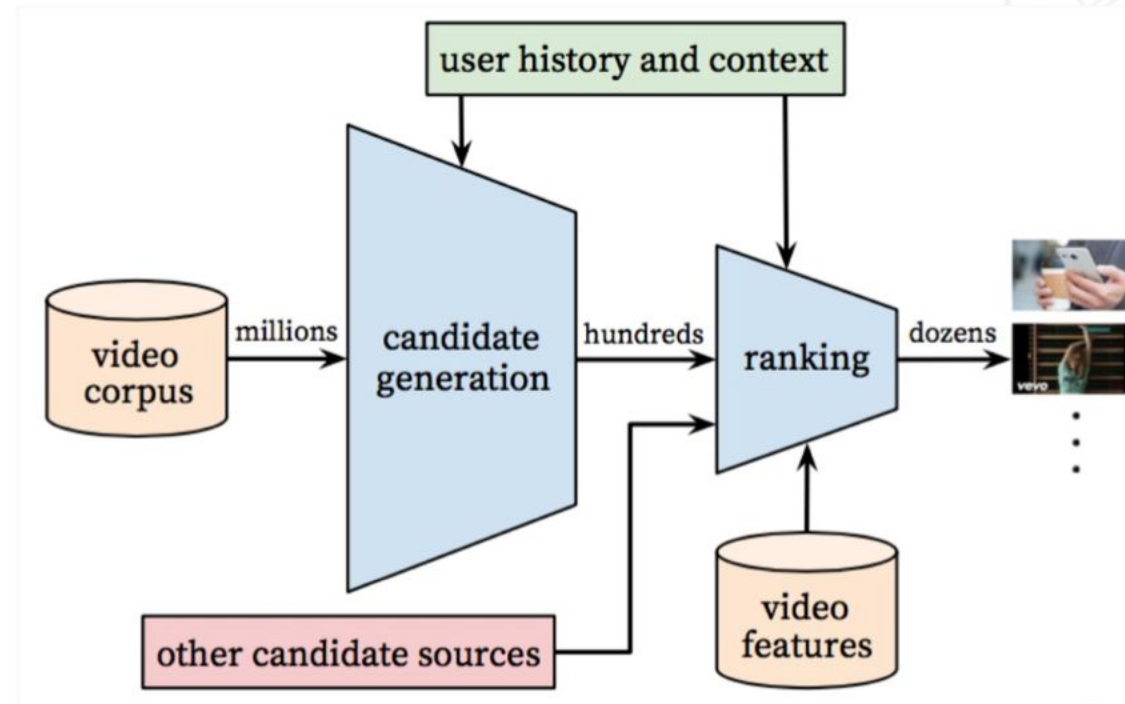
Figure 4: Wide & Deep model structure for apps recommendation.

Table 1: Offline & online metrics of different models. Online Acquisition Gain is relative to the control.

Model	Offline AUC	Online Acquisition Gain
Wide (control)	0.726	0%
Deep	0.722	+2.9%
Wide & Deep	0.728	+3.9%

(Un) Sistema Recomendador de YouTube

- El Sistema Recomendador de YouTube (2016)

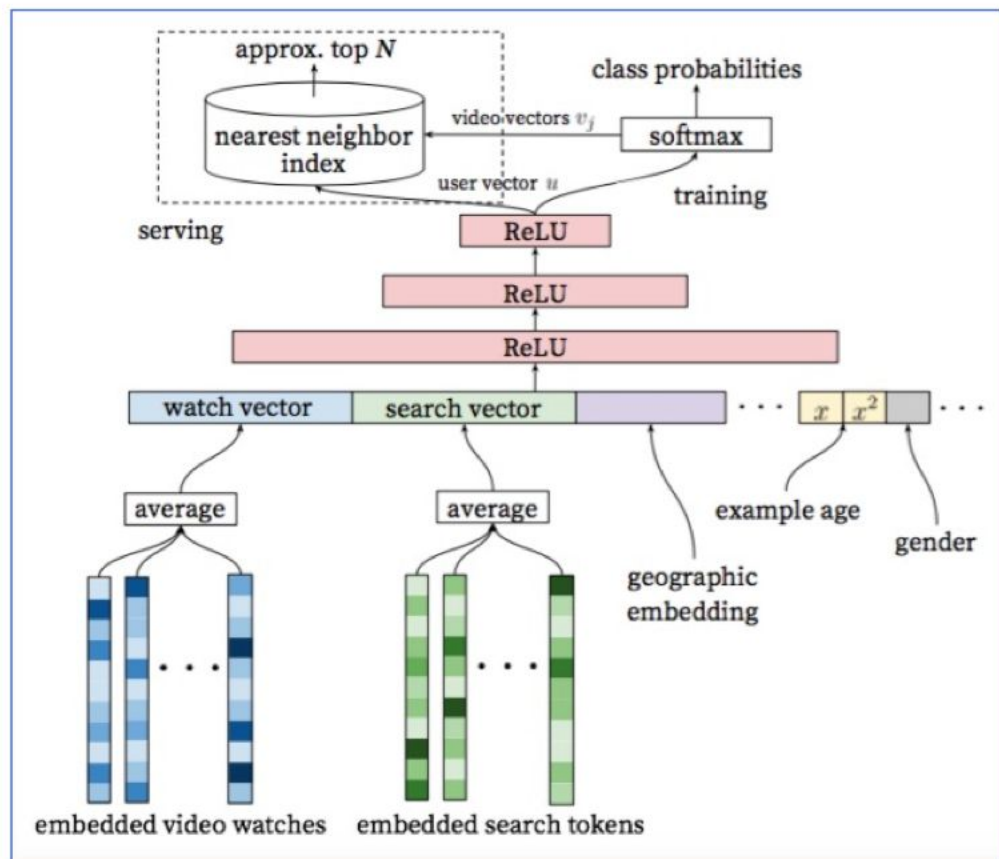


SE DIVIDE EN DOS
GRANDES MÓDULOS:

1. GENERADOR DE CANDIDATOS.
2. RANKING TOP N.

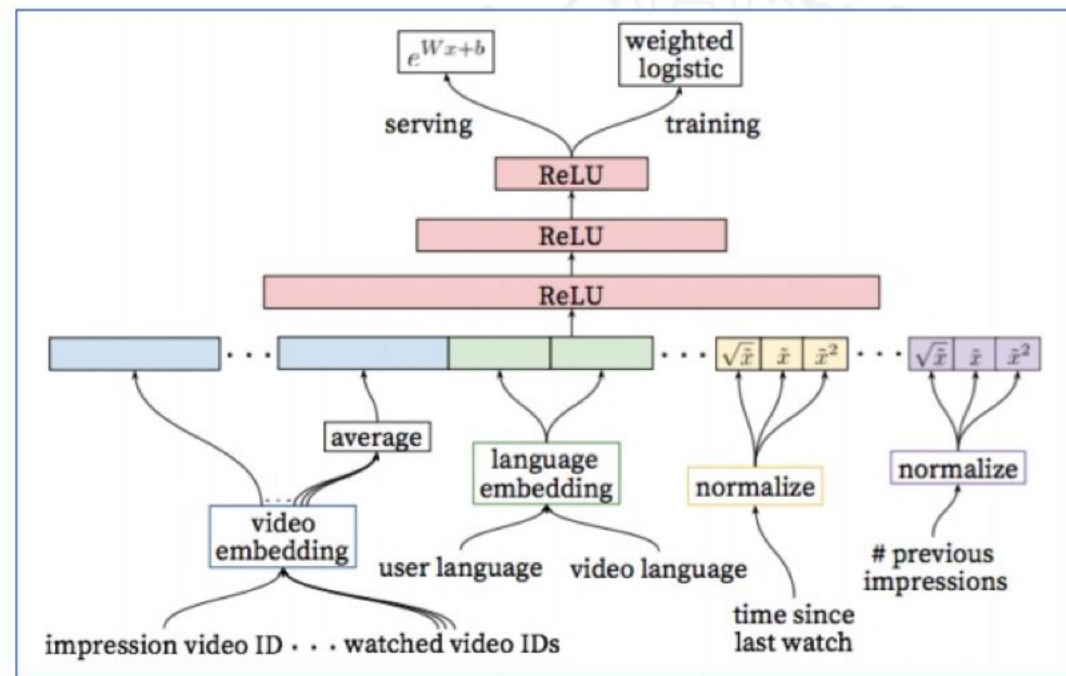
Covington, P., Adams, J., & Sargin, E. (2016). Deep neural networks for YouTube recommendations. In *Proceedings of the 10th ACM conference on recommender systems* (pp. 191-198). ACM.

Redes Neuronales en YouTube (2016-19)



Generación de Candidatos

INFORMACIÓN GENERAL



Ranking

INFORMACIÓN ESPECÍFICA (+ DIFÍCIL DE
COMPUTAR)

Spotify

- Spotify combina varias técnicas:
 - Filtrado Colaborativo: Si a gente parecida a ti le gusta Pink Floyd, a ti te gustará Pink Floyd ...
 - Contenido: Spotify compró hace unos años EchoNest

<https://developer.spotify.com/documentation/web-api/reference/tracks/get-several-audio-features/>

```
{ "audio_features":  
  [ { "danceability": 0.808,  
      "energy": 0.626,  
      "key": 7,  
      "loudness": -12.733,  
      "mode": 1,  
      "speechiness": 0.168,  
      "acousticness": 0.00187,  
      "instrumentalness": 0.159,  
      "liveness": 0.376,  
      "valence": 0.369,  
      "tempo": 123.99,  
      "type": "audio_features",  
      "id": "4JpKVNYnVcJ8tuMKjAj50A",  
      "uri":  
"spotify:track:4JpKVNYnVcJ8tuMKjAj50A",  
      "track_href":
```

Spotify: User Profile

Listening Profile for **Brianna**

Joined Spotify: Feb 2013

Top Tracks

Past 4 Weeks

		# of plays
1. Box Fan Sound	Tinsoft's White Noise Sleep Sounds	43
2. In The River - Live	Jesus Culture	7
3. Síguelo Bailando	Ozuna	7

All Time

1. Box Fan Sound	Tinsoft's White Noise Sleep Sounds	61
2. The Middle	Zedd, Maren Morris, Grey	59
3. Your Body Is a Wonderland	John Mayer	57

Top Artists

	# of plays
1. Chance the Rapper	44
2. Lizzo	23
3. Ozuna	18

1. John Mayer	1047
2. Justin Timberlake	523
3. Ariana Grande	445

In 2018, you listened to **694** hours of music, **half an hour** of podcasts, and watched **2.75** hours of videos.
So far in 2019, you've listened to **573** hours of music, **40 minutes** podcasts, and watched **half an hour** of videos.

You added **1441** songs to **23** personal playlists, **331** songs to **2** collaborative playlists.

In 2018, you listened to **694** hours of music, half an hour of podcasts, and watched **2.75** hours of videos.
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Top Genres: pop, dance pop, post-teen pop, r&b, folk-pop

Most Diverse Genres: indie anthem-folk / trap music

Top Era: You listen to more artists from the **2010s** than any other decade.

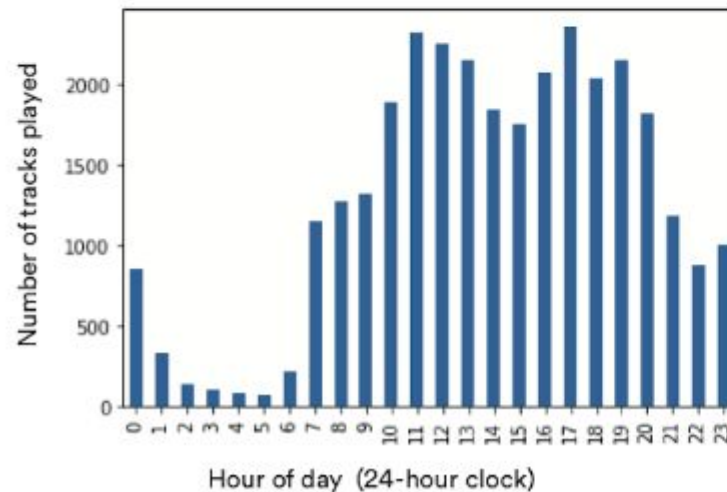
Top City: You listen to more artists from **Nashville** than from any other U.S. city.

Top Country: Outside the U.S., you listen to artists from **France, Germany, Ireland, Canada and the UK** equally.

Demographics: Based on your listening we estimate that you're a **29** year old female.

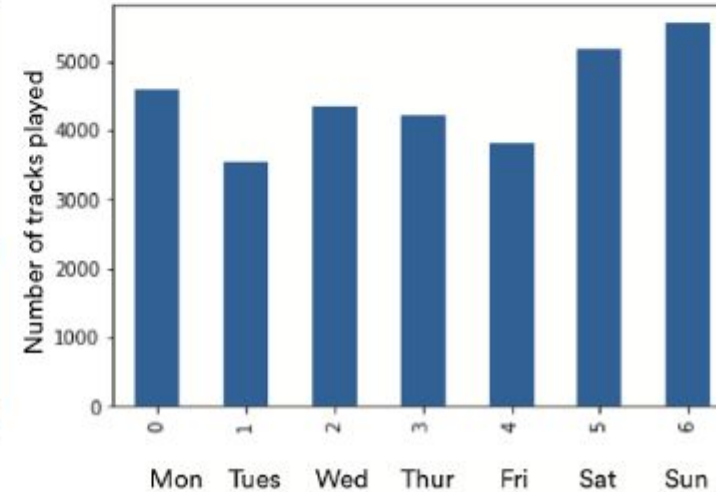
Listening by Time of Day

Top Hour: 5 pm

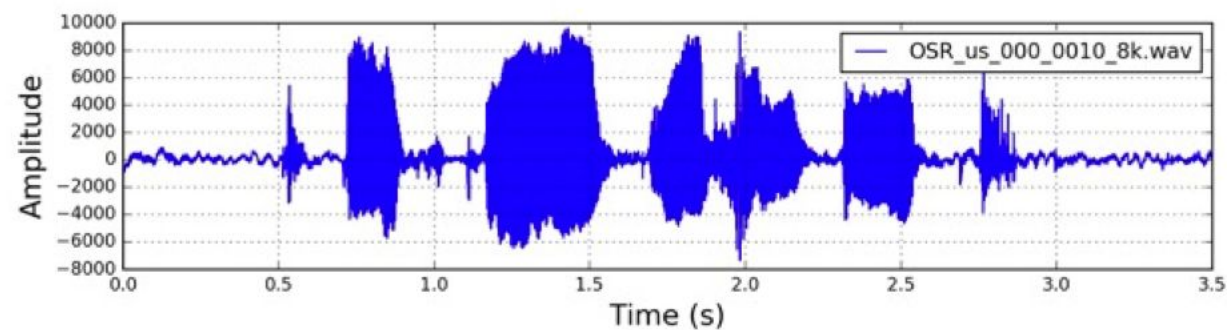


Listening by Day of Week

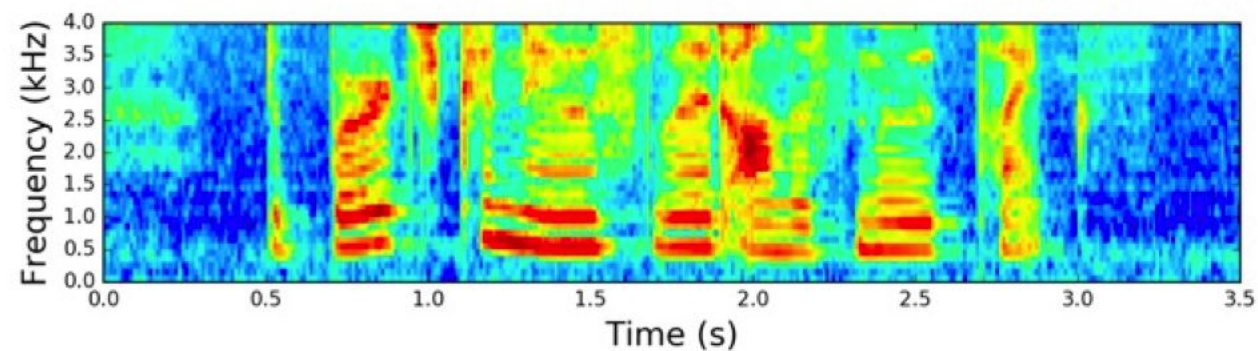
Top Day: Sunday



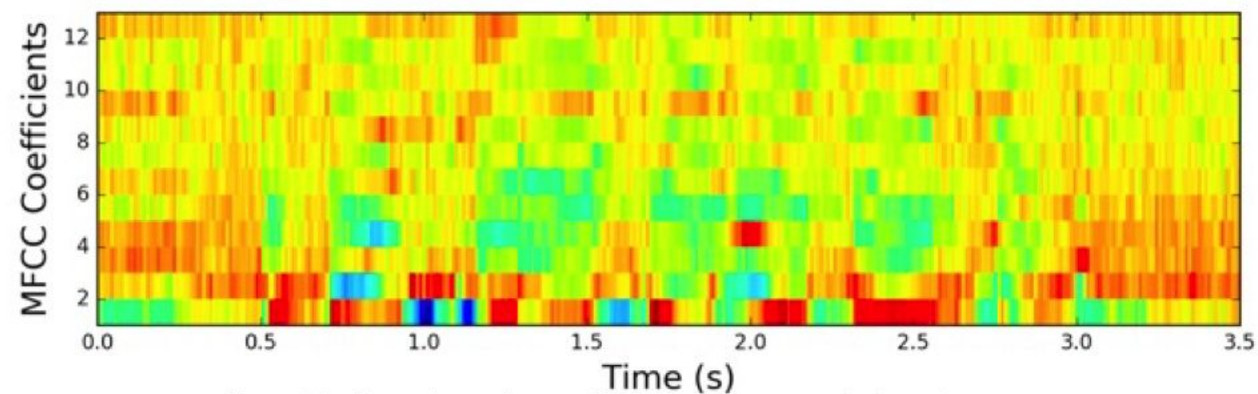
MFCC



Señal de audio

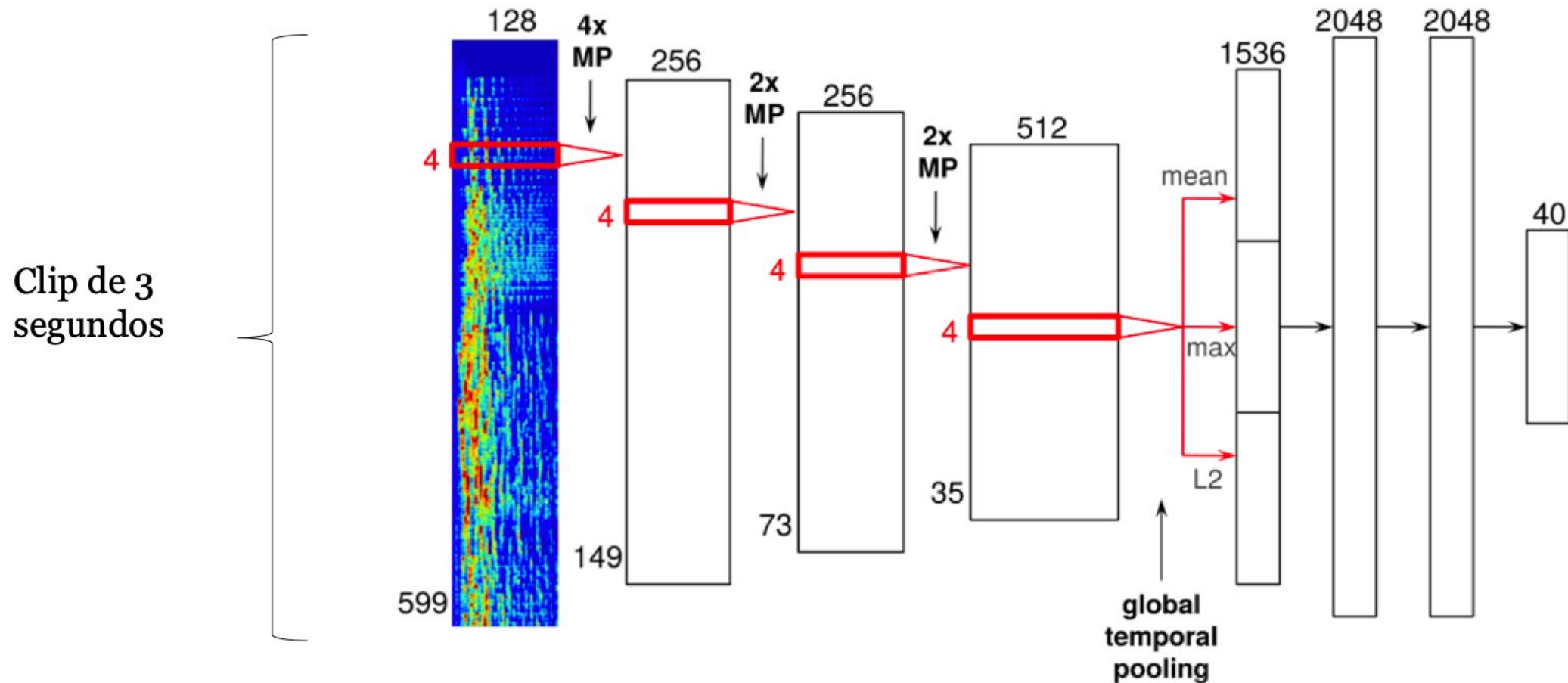


Espectrograma

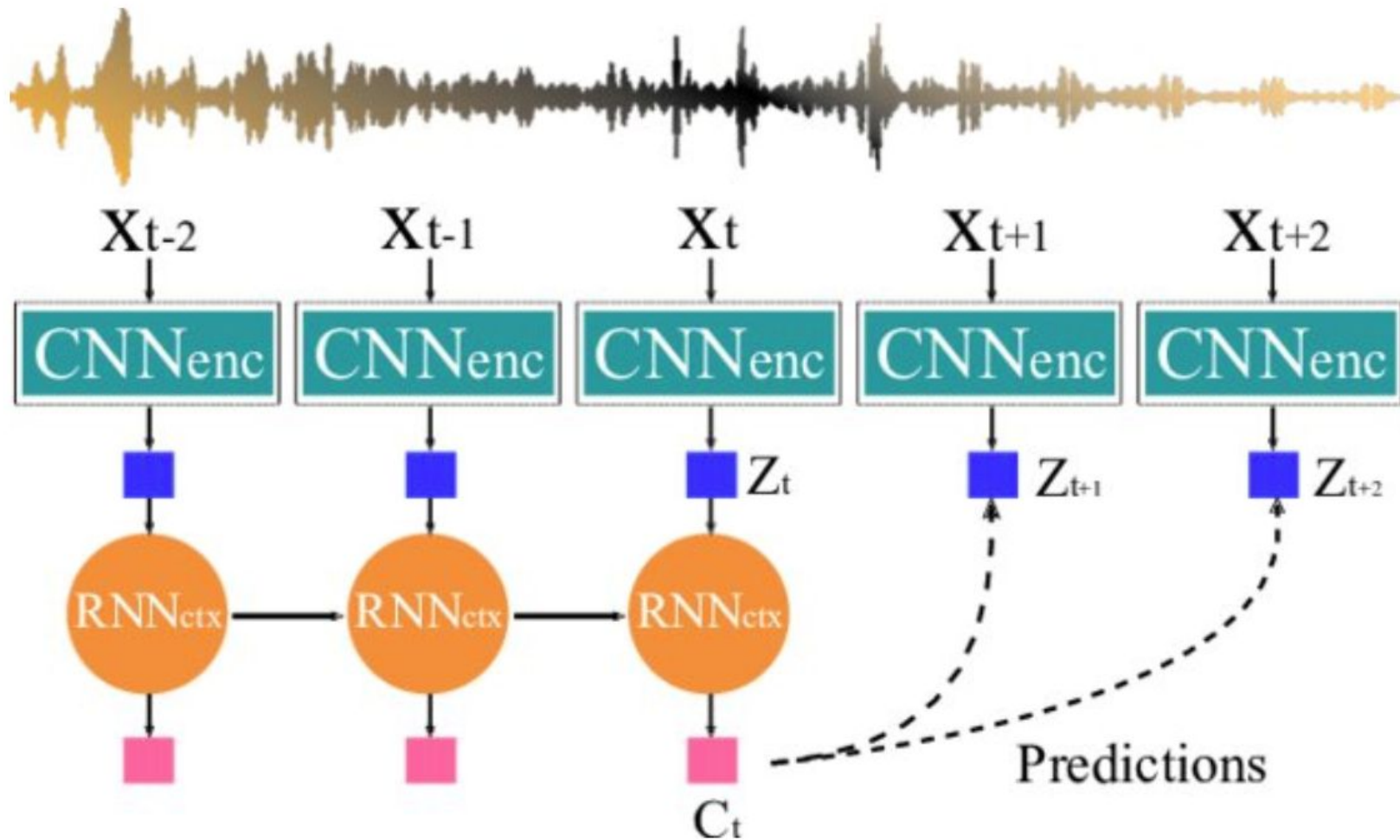


MFCC

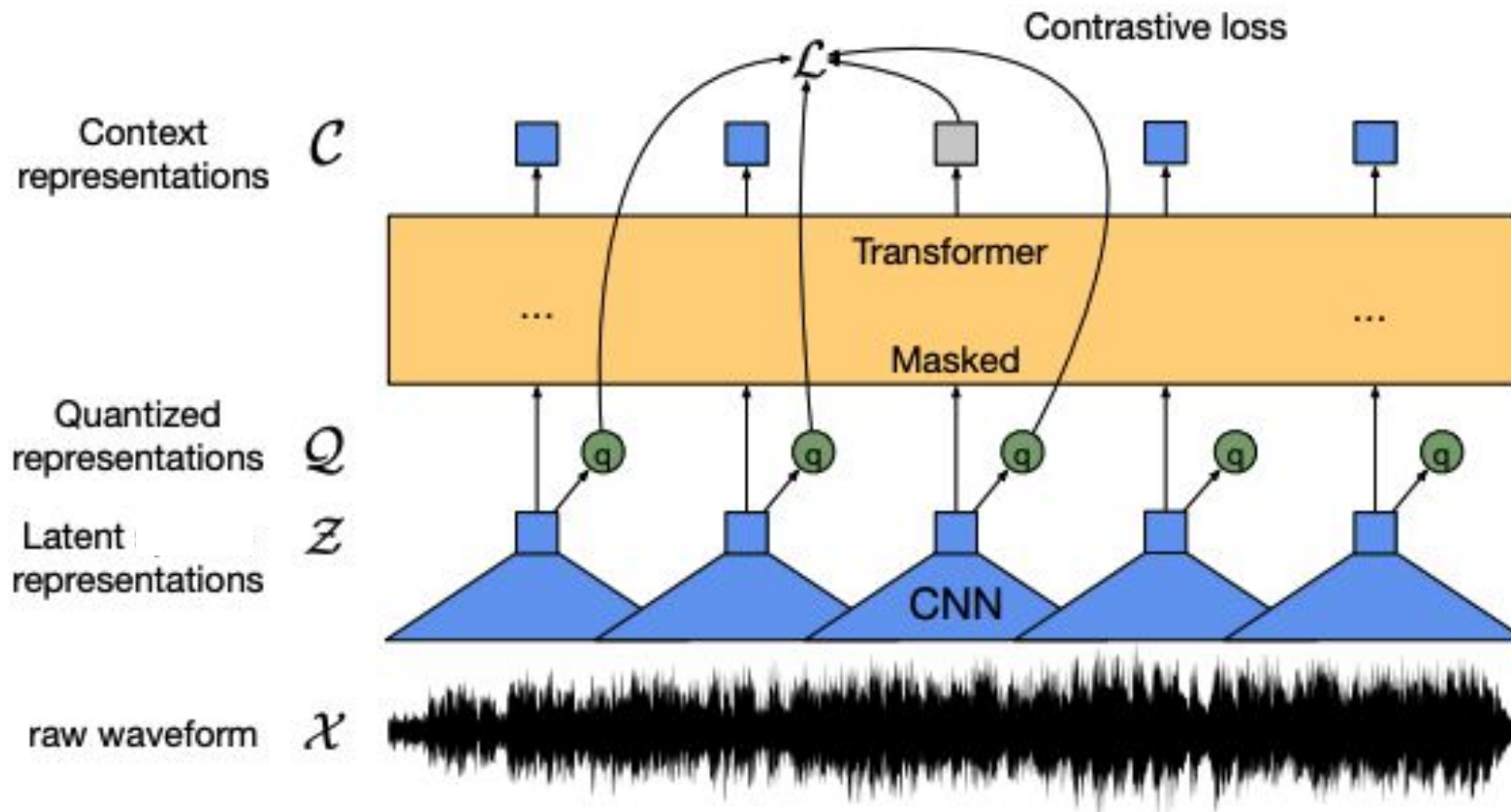
DL en Van den Oord et al (2013)



wav2vec RNN



transformers wav2vec: convertir audio a embedding.



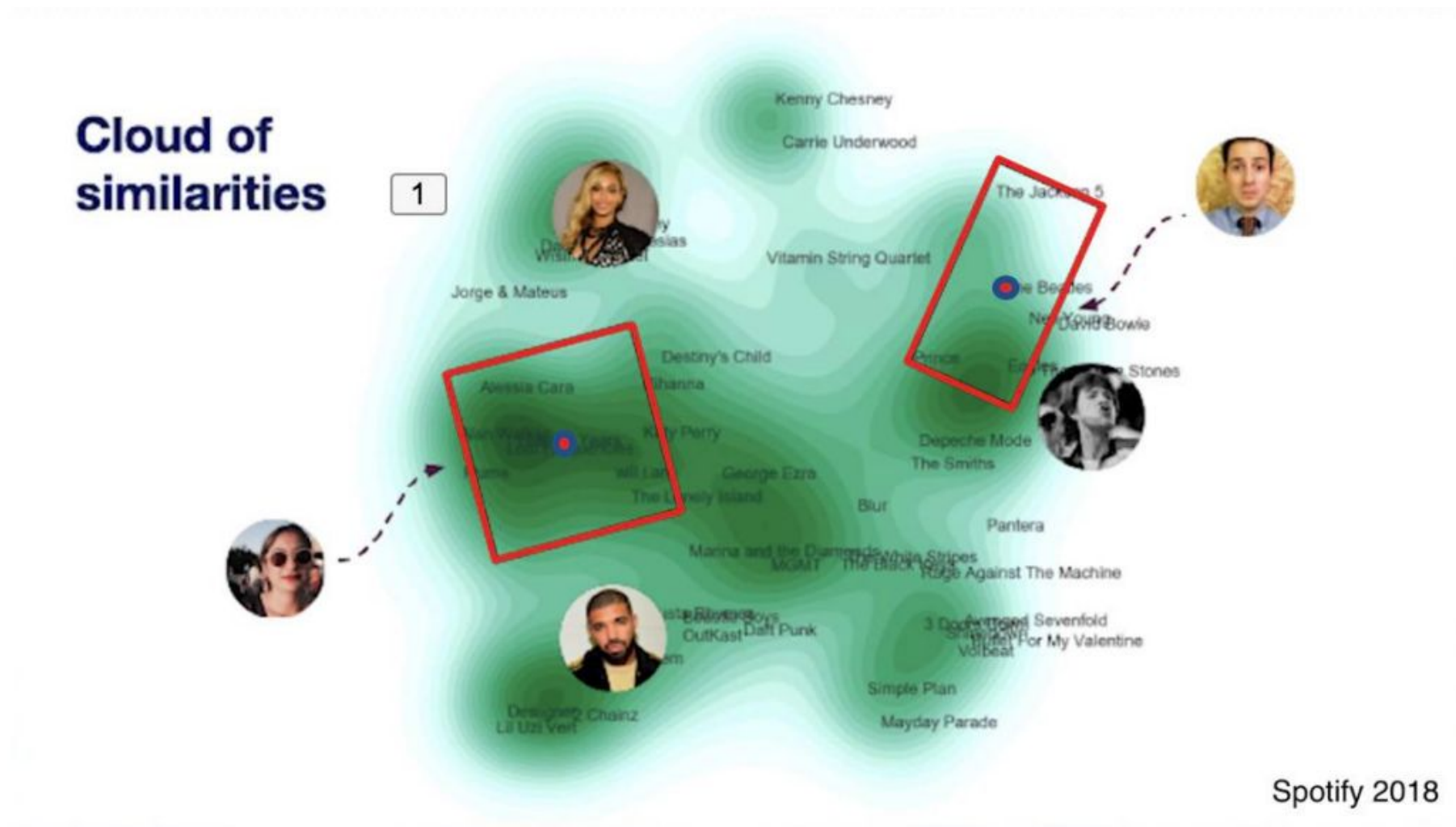


Figure 2-1: Music Recommendations Via Vectors at Spotify

VisRank & CuratorNet

- Publications:

- Messina, P., Cartagena, M., Cerda, P., del Rio, F., & Parra, D. (2020). CuratorNet: Visually-aware Recommendation of Art Images. arXiv preprint arXiv:2009.04426. *Proceedings of ComplexRec-ImpactRS workshop, co-located at RecSys 2020*
- Messina, P., Dominguez, V., Parra, D., Trattner, C., & Soto, A. (2019). Content-based artwork recommendation: integrating painting metadata with neural and manually-engineered visual features. *User Modeling and User-Adapted Interaction*, 29(2), 251-290.

Context: Ugallery

- Online art store
- Focuses specially in emergent artists: to help them sell their paintings
- Our partners in this project

UGALLERY

original art. original you.

FREE SHIPPING AND RETURNS!

TOP ARTISTS STAFF FAVORITES NEW ART COLLECTIONS BLOG Customer Login Artist Login

MEDIUM STYLE SUBJECT SIZE | REGISTRY COMMISSIONS FOR THE TRADE

2334 PIECES

Orientation

☐ Horizontal (1012)

☐ Square (678)

☐ Vertical (644)

Size

Height: 0"- 60"+

Width: 0"- 60"+

Medium

☒ Oil Painting (2334)

☐ Acrylic Painting (2026)

☐ Photography (1372)

☐ Mixed Media Artwork (922)

☐ Watercolor Painting (308)

☐ Sculpture (117)

☐ Printmaking (108)

☐ Digital Printmaking (73)

☐ Drawing Artwork (64)

☐ Encaustic Artwork (56)

Style

Color

Oil Painting

Sort By ▼



Mitchell Freifeld
25" x 30", oil painting
Elephant Rock Motel: \$1100



Morgan Fite
30" x 40", oil painting
Revive: \$1800



Onello Marrero
11" x 14", oil painting
Feeding Time: \$575



Faith Taylor
20" x 24", oil painting
Exhale: \$1100



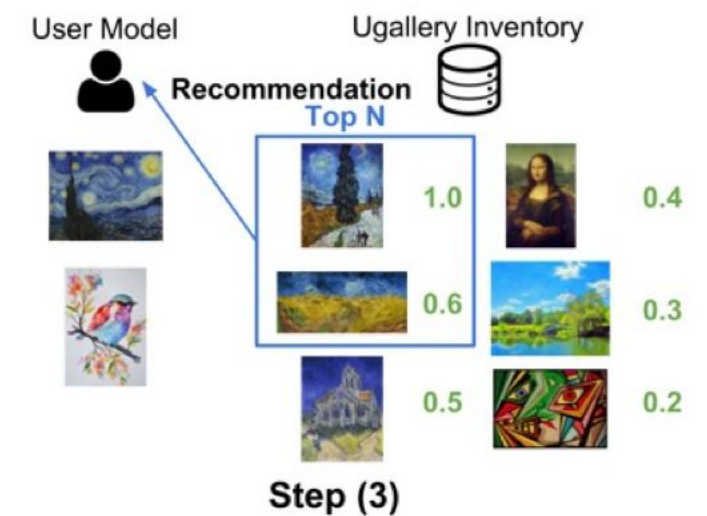
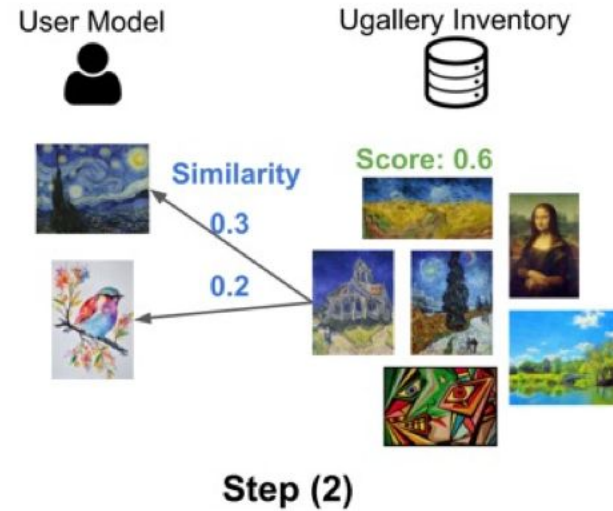
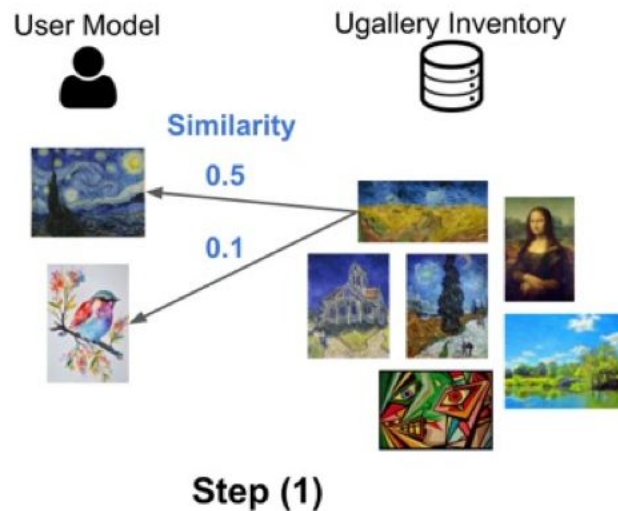
Suren Nersisyan
24" x 30", oil painting
Midnight in the City: \$700



Gail Greene
22" x 28", oil painting
Anticipation: \$1025

Content-based Recommendation

- User Transactions on mostly one-of-a-kind paintings, difficult to do collaborative filtering



Transfer Learning and fine-tuning

- del Rio, F., Messina, P., Dominguez, V., & Parra, D. (2018). Do Better ImageNet Models Transfer Better... for Image Recommendation?. *arXiv preprint arXiv:1807.09870*.
- ResNet was the best for transfer learning (with and w/o finetuning)

CNN	Artwork Image Recommendation				ILSVRC-2012-CLS	
	R@20	P@20	MRR@20	nDCG@20	Top-1 Acc. (%)	Top-5 Acc. (%)
ResNet50	.1632	.0141	.0979	.1253	75.2	92.2
VGG19	.1398	.0124	.0750	.1008	71.1	89.8
NASNet Large	.1379	.0120	.0743	.0998	82.7	96.2
InceptionV3	.1332	.0125	.0744	.1007	78.0	93.9
InceptionResNetV2	.1302	.0117	.0692	.0936	80.4	95.3
Random	.0172	.0013	.0051	.0093	-	-

Table 1: Results of different pre-trained embeddings at the artwork image recommendation task to the left (R:Recall, P:Precision), and their performance at the ILSVRC Challenge trained on ImageNet dataset (Acc: Accuracy). The top methods in both tasks do not correlate.

CuratorNet

- Inspired by both VBPR (2015) and YouTube Recsys (2016)

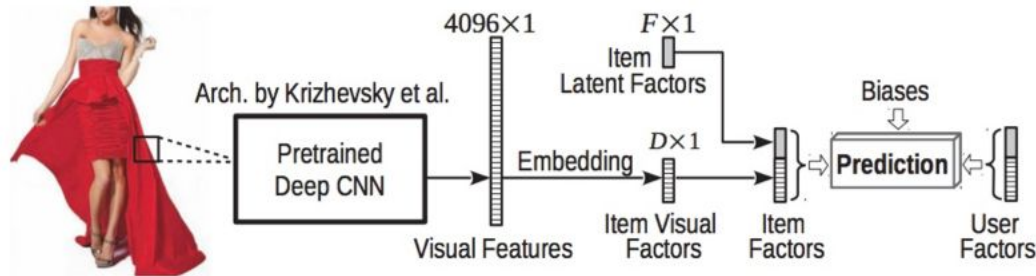
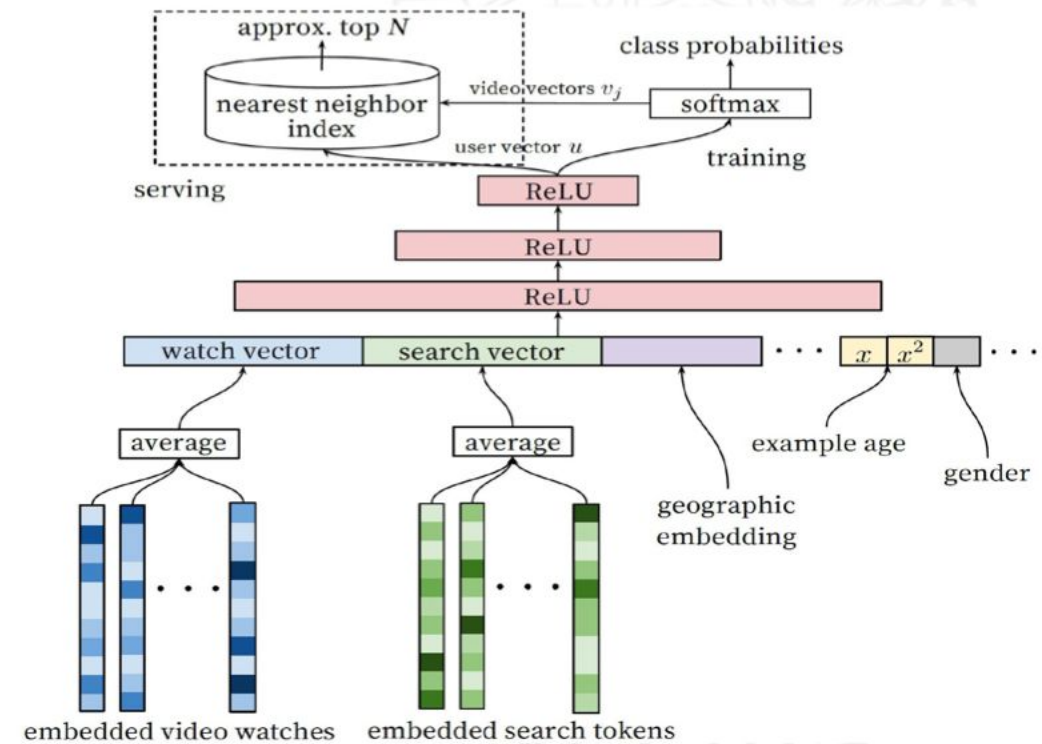


Figure 1: Diagram of our preference predictor. Rating dimensions consist of visual factors and latent (non-visual) factors. Inner products between users and item factors model the compatibility between users and items.

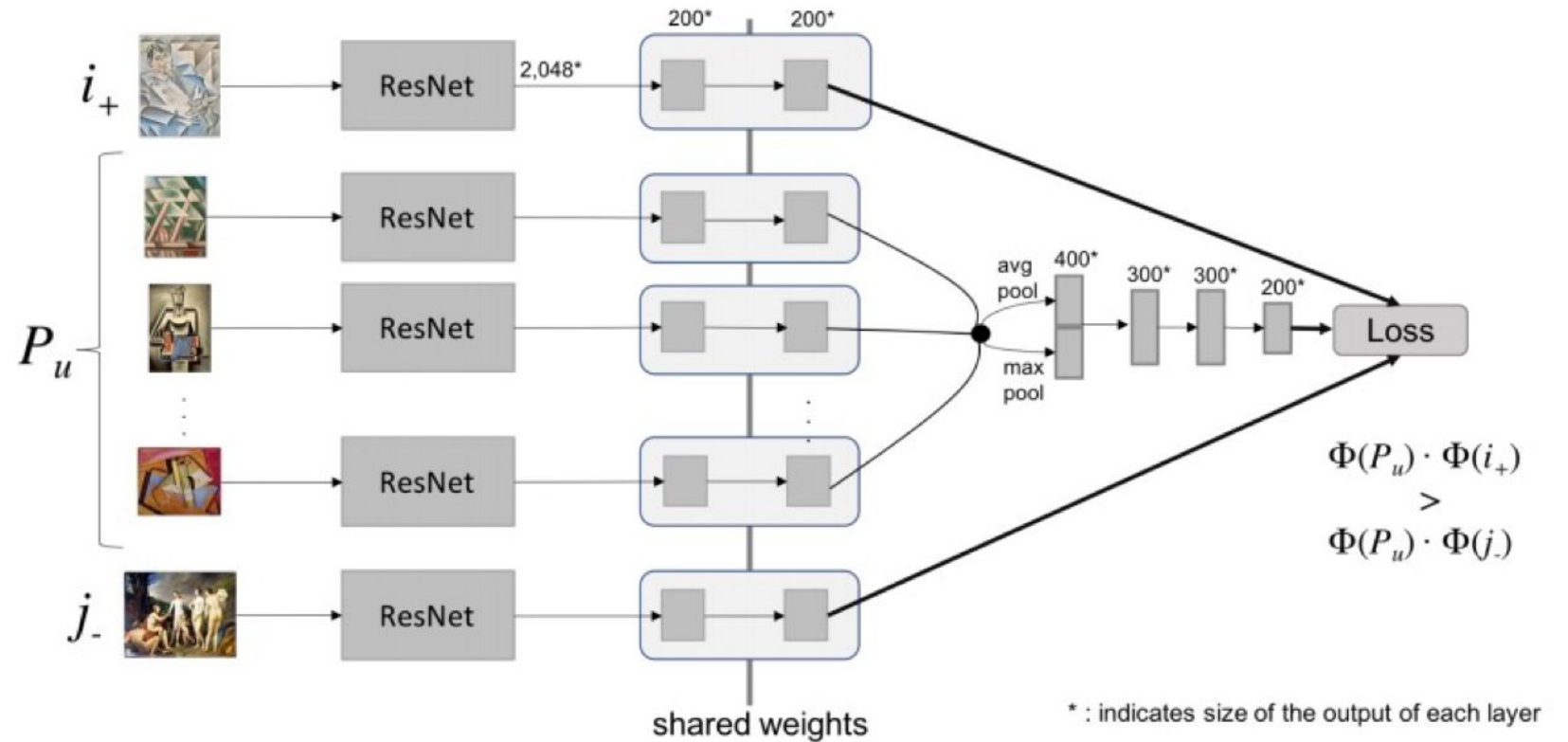
He, R., & McAuley, J. (2016). VBPR: visual Bayesian Personalized Ranking from implicit feedback. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence* (pp. 144-150).



Covington, P., Adams, J., & Sargin, E. (2016, September). Deep neural networks for youtube recommendations. In *Proceedings of the 10th ACM conference on recommender systems* (pp. 191-198).

CuratorNet

- Using BPR framework, we learn from triples (P_u, i_+, j_-)



Messina, P., Cartagena, M., Cerda, P., del Rio, F., & Parra, D. (2020). CuratorNet: Visually-aware Recommendation of Art Images. Proceedings of ComplexRec-ImpactRS workshop, co-located at RecSys 2020

VisRank (baseline)

$$s(u,i) = \text{score}(\text{ContentBasedProfile}(u), \text{Content}(i))$$

User Model



Ugallery Inventory



Similarity



0.5

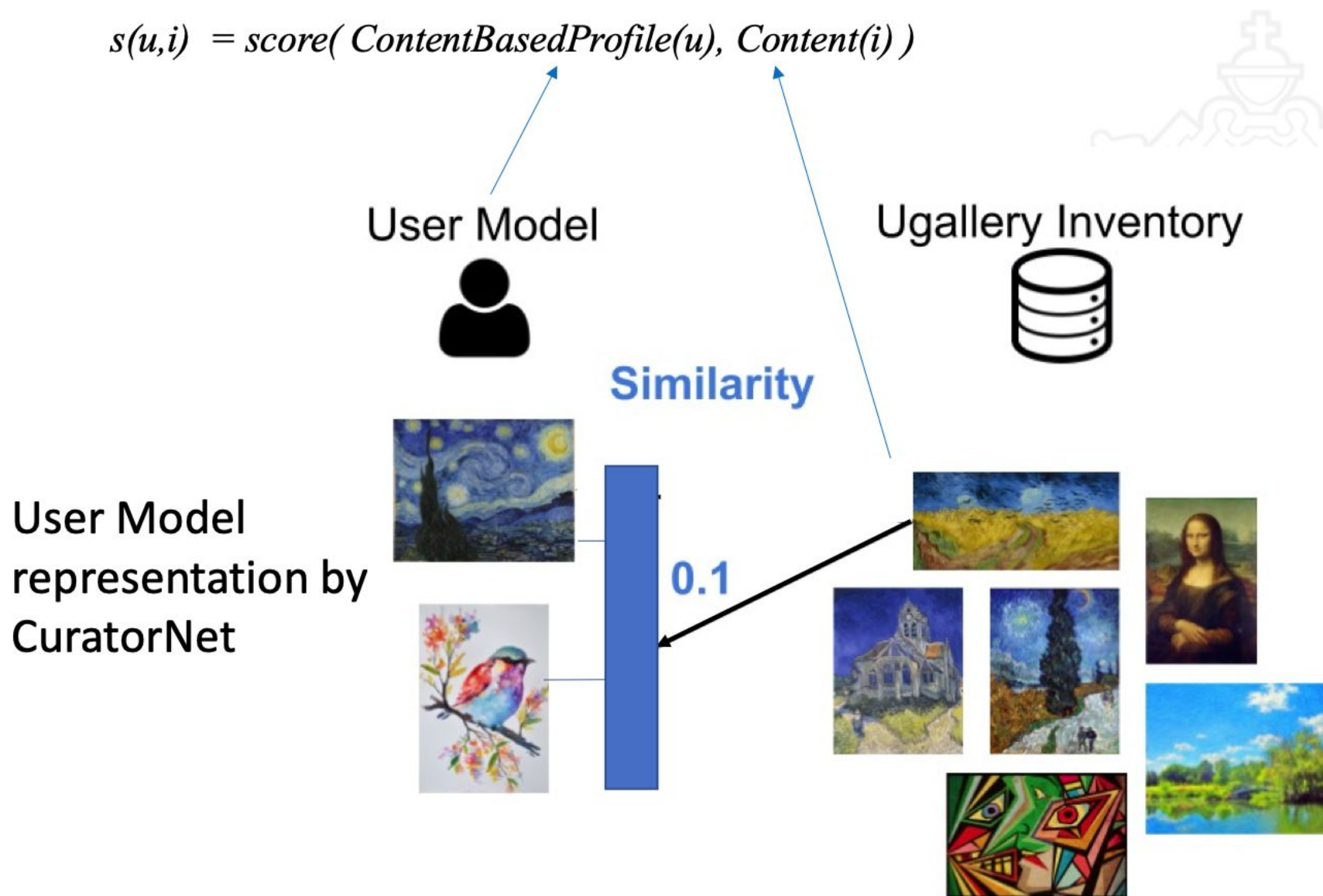


0.1



CuratorNet

$$s(u,i) = \text{score}(\text{ContentBasedProfile}(u), \text{Content}(i))$$



Results: VisRank, VBPR, CuratorNet



Method	λ (L2 Reg.)	AUC	R@20	P@20	nDCG@20	R@100	P@100	nDCG@100
Oracle	–	1.0000	1.0000	.0655	1.0000	1.0000	.0131	1.0000
CuratorNet	.0001	.7204	.1683	.0106	.0966	.3200	.0040	.1246
CuratorNet	.001	.7177	.1566	.0094	.0895	.2937	.0037	.1160
VisRank	–	.7151	.1521	.0093	.0956	.2765	.0034	.1195
CuratorNet	0	.7131	.1689	.0100	.0977	.3048	.0038	.1239
CuratorNet	.01	.7125	.1235	.0075	.0635	.2548	.0032	.0904
VBPR	.0001	.6641	.1368	.0081	.0728	.2399	.0030	.0923
VBPR	0	.6543	.1287	.0078	.0670	.2077	.0026	.0829
VBPR	.001	.6410	.0830	.0047	.0387	.1948	.0024	.0620
VBPR	.01	.5489	.0101	.0005	.0039	.0506	.0006	.0118
Random	–	.4973	.0103	.0006	.0041	.0322	.0005	.0098

Conclusion

- CuratorNet improves upon VBPR and VisRank for the case of one-of-a-kind recommendation
- Unlike VBPR, CuratorNet does not need re-training to recommend to new users since it does not explicitly train user factors
- The proposed sampling guidelines benefit both CuratorNet and VBPR

Code and Experimental Data <https://github.com/ialab-puc/CuratorNet>

Ecosistema de DeepLearning

- <https://www.tensorflow.org/recommenders>
- <https://pytorch.org/blog/introducing-torchrec/>



PRÁCTICO

https://colab.research.google.com/drive/1b_RCuRsSgl_Xz22OldHtTTew7zVMm9gn?usp=sharing

